

Administrator Guide

Administrator Guide: Open Build Service

by Karsten Keil

Publication Date: 16 Oct 2026

https://documentation.suse.com

Copyright © 2016

Copyright © 2006– 2026 SUSE LLC and contributors. All rights reserved.

Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.2 or (at your option) version 1.3; with the Invariant Section being this copyright notice and license. A copy of the license version 1.2 is included in the section entitled “GNU Free Documentation License”.

For SUSE trademarks, see http://www.suse.com/company/legal/. All other third-party trademarks are the property of their respective owners. Trademark symbols (®, ™ etc.) denote trademarks of SUSE and its affiliates. Asterisks (*) denote third-party trademarks.

All information found in this book has been compiled with utmost attention to detail. However, this does not guarantee complete accuracy. Neither SUSE LLC, its affiliates, the authors nor the translators shall be held liable for possible errors or the consequences thereof.

Contents

About this Guide viii

1 Available Documentation viii

2 Feedback viii

3 Documentation Conventions ix

4 Contributing to the Documentation x

1 High-level Overview 1

1.1 OBS Components 1

Front-end 2 • Back-end 2 • Command Line Client 4 • Content Delivery Server 4 • Requirements 4

1.2 OBS Appliances 5

Server Appliance 5 • Worker Appliance 5 • Image Types 5 • Deployment 6 • Separating Data from the System 6 • Updating the Appliance 7

1.3 Back-end Administration 7

Services 7 • Advanced Setups 8

1.4 Front-end Administration 9

Delayed Jobs 9 • Full Text Search 9

2 Installation and Configuration 10

2.1 Planning 10

Resource Planning 10

2.2 Simple Installation 11

Back-end Installation 11 • Front-end Installation 15 • Online Configuration 18

2.3 Worker Farm 21

2.4 Distributed Setup 22

2.5 Monitoring 25

Endpoint Checks 25 • Common Checks 26 • Other Checks 27

3 File System Overview 29

3.1 Configuration Files 29

Front-end Configuration 29 • Back-end Configuration 38

3.2 Log Files 58

Front-end 58 • Back-end 58

3.3 `/srv/obs` Tree 59

`build` Directory 59 • `cloudupload` Directory 60 • `db` Directory 60 • `diffcache` Directory 61 • `events` Directory 61 • `info` Directory 61 • `jobs` Directory 61 • `log` Directory 61 • `projects` Directory 61 • `remotecache` Directory 61 • `repos` Directory 62 • `repos_sync` Directory 62 • `run` Directory 62 • `sources` Directory 62 • `trees` Directory 62 • `upload` Directory 62 • `workers` Directory 63

3.4 Metadata 63

OBS Revision Control 63 • Project Metadata 64 • Package Metadata 66 • Attribute Metadata 66 • Job Files 67

4 Security Concepts 69

4.1 General Paradigm 69

Frontend 69 • Build Environment 70 • Source Revision System 71 • Permission Handling 71 • Signature Handling 71

4.2 Trust Zones 72

Public Zones 72 • Demilitarized Zone (DMZ) 73 • Internal Zone 73 • Worker Zone 74 • Signing Server 74

5 Administration 76

5.1 Tools 76

`obs_admin` 76 • **`osc`** 79

5.2 Managing Build Targets 80
Interconnect 80 • Importing Distributions 81

5.3 Source Services 81
Using Services for Validation 82 • Different Modes When Using Services 82 • Storage of Source Service Definitions 84 • Dropping a Source Service Again 84

5.4 Source Publisher 84
Configuring Source Publisher 85 • Considerations 85

5.5 Dispatch Priorities 85
The `/build/_dispatchprios` API Call 86 • `dispatch_adjust` Array 87

5.6 Publisher Hooks 88
Configuring Publisher Hooks 88 • Example Publisher Scripts 90

5.7 Unpublisher Hooks 91
Configuring Unpublisher Hooks 92 • Example Unpublisher Scripts 93

5.8 Managing Users and Groups 95
User and Group Roles 95 • Standalone User and Group Database 95 • Users and Group Maintainers 96 • Gravatar for Groups 96 • Proxy Mode 96 • LDAP/Active Directory 97 • Authentication Methods 103

5.9 Message Bus for Event Notifications 105
RabbitMQ 106

5.10 Backup 114
Places to consider 114 • Backup strategies 115

5.11 Restore 116
Check and repair database inconsistencies 116 • Binaries 117

5.12 Repair Data Corruption 117

5.13 Spider Identification 117

5.14 Worker in Kubernetes 119

6 Troubleshooting 124

6.1 General Hints 124

6.2 Debugging Front-end Problems 125

7 Setting Up a Local OBS Instance 126

7.1 Testing OBS on Microsoft Windows Using VMware Player 126

7.2 Installing a Readymade OBS Appliance in a VirtualBox 127

7.3 First Steps with Your New OBS Server 128

A GNU Licenses 130

About this Guide

This guide is part of the Open Build Service documentation. These books are considered to contain only reviewed content, establishing the reference documentation of OBS.

This guide does not focus on a specific OBS version. It is also not a replacement of the documentation inside of the openSUSE Wiki (https://en.opensuse.org/Portal:Build_Service). However, content from the wiki may be included in these books in a consolidated form.

1 Available Documentation

The following documentation is available for OBS:

Book "Administrator Guide"

This guide offers information about the initial setup and maintenance for running Open Build Service instances.

Book "User Guide"

This guide is intended for users of Open Build Service. The first part describes basic workflows for working with packages on Open Build Service. This includes checking out a package from an upstream project, creating patches, branching a repository, and more. The following parts go into more detail and contain information on backgrounds, setting up your computer for working with OBS, and usage scenarios. The *Best Practices* part offers step-by-step instructions for the most common features of the Open Build Service and the openSUSE Build Service. The last part covers ideas and motivations, concepts and processes of the Open Build Service.

2 Feedback

Several feedback channels are available:

Bugs and Enhancement Requests

Help for openSUSE is provided by the community. Refer to https://en.opensuse.org/Portal:Support for more information.

Bug Reports

To report bugs for Open Build Service, go to https://bugzilla.opensuse.org/, log in, and click *New*.

Mail

For feedback on the documentation of this product, you can also send a mail to `doc-team@suse.com`. Make sure to include the document title, the product version and the publication date of the documentation. To report errors or suggest enhancements, provide a concise description of the problem and refer to the respective section number and page (or URL).

3 Documentation Conventions

The following notices and typographical conventions are used in this documentation:

- `/etc/passwd`: directory names and file names
- *`PLACEHOLDER`*: replace *`PLACEHOLDER`* with the actual value
- `PATH`: the environment variable PATH
- **`ls`**, `--help`: commands, options, and parameters
- `user`: users or groups
- `package name` : name of a package
- `Alt` , `Alt` – `F1` : a key to press or a key combination; keys are shown in uppercase as on a keyboard
- *File*, *File* › *Save As*: menu items, buttons
- *Dancing Penguins* (Chapter *Penguins*, ↑Another Manual): This is a reference to a chapter in another manual.
- Commands that must be run with `root` privileges. Often you can also prefix these commands with the **`sudo`** command to run them as non-privileged user.

```
root # command
geeko > sudo command
```

- Commands that can be run by non-privileged users.

```
geeko > command
```

- Notices



Warning: Warning Notice

Vital information you must be aware of before proceeding. Warns you about security issues, potential loss of data, damage to hardware, or physical hazards.



Important: Important Notice

Important information you should be aware of before proceeding.



Note: Note Notice

Additional information, for example about differences in software versions.



Tip: Tip Notice

Helpful information, like a guideline or a piece of practical advice.

4 Contributing to the Documentation

The OBS documentation is written by the community. And you can help too!

Especially as an advanced user or an administrator of OBS, there will be many topics where you can pitch in even if your English is not the most polished. Conversely, if you are not very experienced with OBS but your English is good: We rely on community editors to improve the language.

This guide is written in DocBook XML which can be converted to HTML or PDF documentation.

To clone the source of this guide, use Git:

```
git clone https://github.com/openSUSE/obs-docu.git
```

To learn how to validate and generate the OBS documentation, see the file `README`.

To submit changes, use GitHub pull requests:

1. Fork your own copy of the repository.
2. Commit your changes into the forked repository.

3. Create a pull request. This can be done at https://github.com/openSUSE/obs-docu.

It is even possible to host instance-specific content in the official Git repository, but it needs to be tagged correctly. For example, parts of this documentation are tagged as `<para os="opensuse">`. In this case, the paragraph will only become visible when creating the openSUSE version of a guide.

1 High-level Overview

This chapter describes the components of an OBS installation and the typical administration tasks for an OBS administrator.

This chapter is not intended to describe special installation hints for a certain OBS version. Refer to the OBS download page (http://openbuildservice.org/download/) for that.

1.1 OBS Components

The OBS is not a monolithic server: it consists of multiple daemons that perform different tasks.

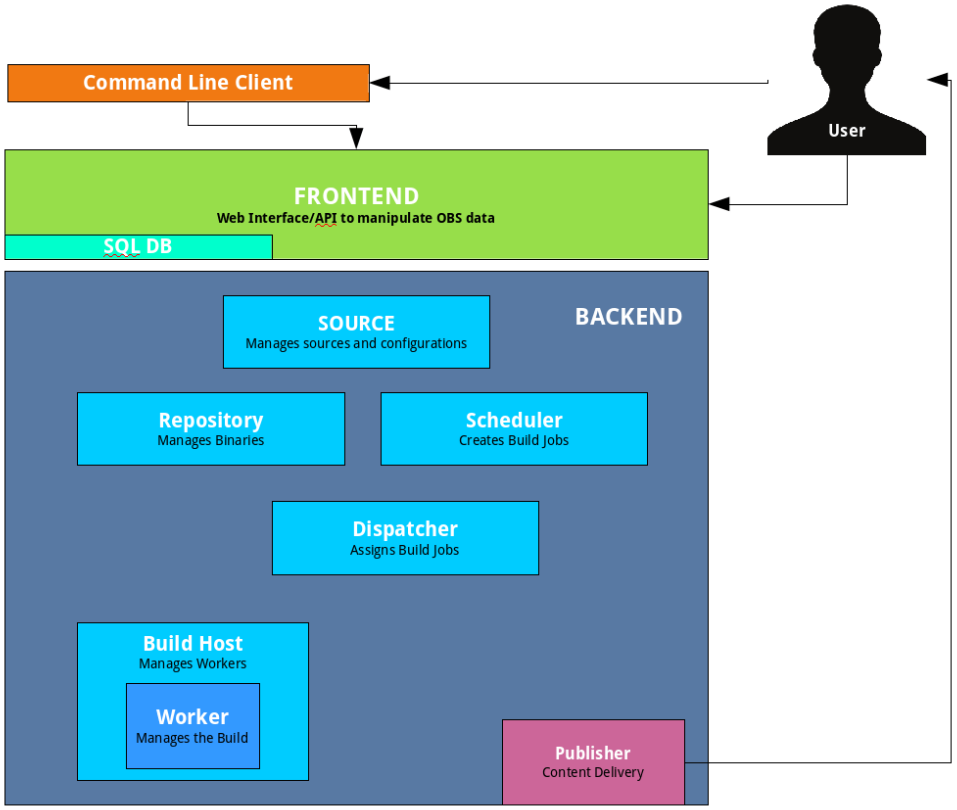


FIGURE 1.1: SIMPLIFIED OBS COMPONENT OVERVIEW

1.1.1 Front-end

The OBS Front-end is a Ruby on Rails application that manages the access and manipulation of OBS data. It provides a web user interface and an application programming interface to do so. Both can be used to create, read, update and delete users, projects, packages, requests and other objects. It also implements additional sub-systems like authentication, search, and email notifications.

1.1.2 Back-end

The OBS Back-end is a collection of Perl applications that manage the source files and build jobs of the OBS.

1.1.2.1 Source Server

Maintains the source repository and project/package configurations. It provides an HTTP interface, which is the only interface to the front-end. It may forward requests to other back-end services. Each OBS installation has exactly one Source Server. It maintains the "sources", "trees" and "projects" directories.

1.1.2.2 Repository Server

A repository server provides access to the binaries via an HTTP interface. It is used by the front-end via the source server only. Workers use the server to register, request the binaries needed for build jobs, and store the results. Notifications for schedulers are also created by repository servers. Each OBS installation has at least one repository server. A larger installation using partitioning has one on each partition.

1.1.2.3 Scheduler

A scheduler calculates the need for build jobs. It detects changes in sources, project configurations or in binaries used in the build environment. It is responsible for starting jobs in the right order and integrating the built binary packages. Each OBS installation has one scheduler per available architecture and partition. It maintains the "build" directory.

1.1.2.4 Dispatcher

The dispatcher takes a job (created by the scheduler) and assigns it to a free worker. It also checks possible build constraints to verify that the worker qualifies for the job. It only notifies a worker about a job; the worker itself downloads the required resources. Each OBS installation has one dispatcher per partition (one of which is the master dispatcher).

1.1.2.5 Publisher

The publisher processes "publish" events from the scheduler for finished repositories. It merges the build result of all architectures into a defined directory structure, creates the required metadata, and optionally syncs it to a download server. It maintains the "repos" directory on the back-end. Each OBS installation has one publisher per partition.

1.1.2.6 Source Publisher

The source publisher processes "sourcepublish" events from the publisher for published binary repositories. It needs to run on the same instance as the source server. It can be used to publish a filesystem structure providing all sources of published binaries. In case of images or containers this also includes the sources of used binary packages.

1.1.2.7 Worker

The workers register with the repository servers. They receive build jobs from the dispatcher. Afterwards they download sources from source server and the required binaries from the repository server(s). They build the package using the build script and send the results back to the repository server. A worker can run on the same host as other services, but most OBS installations have dedicated hardware for the workers.

1.1.2.8 Signer

The signer handles signing events and calls an external tool to execute the signing. Each OBS installation usually has one signer per partition and one on the source server installation.

1.1.2.9 Warden

The warden monitors the workers and detects crashed or hanging workers. Their build jobs will be canceled and restarted on another host. Each OBS installation can have one Warden service running on each partition.

1.1.2.10 Download on Demand Updater (dodup)

The download on demand updater monitors all external repositories which are defined as "download on demand" resources. It polls for changes in the metadata and re-downloads the metadata as needed. The scheduler will be notified to recalculate the build jobs depending on these repositories afterwards. Each OBS installation can have one dodup service running on each partition.

1.1.2.11 Delta Store

The delta store daemon maintains the deltas in the source storage. Multiple obscpio archives can be stored in one deltastore to avoid duplication on disk. This service calculates the delta and maintains the delta store. Each OBS installation can have one delta store process running next to the source server.

1.1.3 Command Line Client

The Open Build Service Commander (osc) is a Python application with a Subversion-style command-line interface. It can be used to manipulate or query data from the OBS through its application programming interface.

1.1.4 Content Delivery Server

The OBS is agnostic about how you serve build results to your users. It will just write repositories to disk. But many people sync these repositories to some content delivery system like MirrorBrain (http://mirrorbrain.org/).

1.1.5 Requirements

We highly recommend, and in fact only test, installations on the SUSE Linux Enterprise Server (https://www.suse.com/products/server/) and openSUSE (http://www.opensuse.org) operating systems. However, there also are installations on Debian and Fedora systems.

The OBS also needs a SQL database (MySQL or MariaDB) for persistent and a memcache daemon for volatile data.

1.2 OBS Appliances

This chapter gives an overview over the different OBS appliances and how to deploy them for production use.

1.2.1 Server Appliance

The OBS server appliance contains a recent openSUSE distribution with a pre-installed and pre-configured OBS front-end, back-end and worker. The operating system on this appliance adapts to the hardware on first boot and defaults to automatic IP and DNS configuration via DHCP.

1.2.2 Worker Appliance

The OBS worker appliance includes a recent openSUSE distribution and the OBS worker component. The operating system on this appliance adapts to the hardware on first boot, defaults to automatic IP and DNS configuration via DHCP and OBS server discovery via SLP.

1.2.3 Image Types

There are different types of OBS appliance images.

TABLE 1.1: APPLIANCE TYPES

File Name Suffix	Appliance for
.vdi	VirtualBox (https://www.virtualbox.org/).
.vmdk	VMware (http://www.vmware.com/) Workstation and Player.

File Name Suffix	Appliance for
	Note Our VirtualBox images are usually better tested.
.qcow2	QEMU/KVM (http://qemu.org).
.raw	Direct writing to a block device
.tgz	Deploying via PXE from a central server

1.2.4 Deployment

To help you deploy the OBS server appliance to a hard disk there is a basic installer that you can boot from a USB stick. The installer can be found on the OBS Download page (http://openbuildservice.org/download/).

The image can be written to a USB stick to boot from it:

```
xzcat obs-server-install.x86_64.raw.xz > /dev/sdX
```

Warning

/dev/sdX is the main device of your USB stick. Do NOT put it into a partition like /dev/sda1. If you use the wrong device, you will destroy all data on it!

How to deploy the other image types deeply depends on your virtualization setup. Describing this is out of scope for this guide, sorry.

1.2.5 Separating Data from the System

For production use you want to separate the OBS data from operating system of the appliance so you can re-deploy the appliance without touching your OBS data. This can be achieved by creating an LVM volume group with the name "OBS". This volume group should be as large as possible because it is getting used by the OBS back-end for data storage and the OBS workers for root/swap/cache file systems. To create an LVM volume prepare a partition of type "8e" and create the LVM via

```
pvcreate /dev/sdX1
vgcreate "OBS" /dev/sdX1
```

Additionally, if the OBS volume group contains a logical volume named “server”, it will be used as the data partition for the server.

```
lvcreate "OBS" -n "server"
mkfs.xfs /dev/OBS/server
```

1.2.6 Updating the Appliance

All images come pre-configured with the right set of repositories and can be updated via the system tools YaST or **`zypper`** at any time. Another way to update is to re-deploy the entire image.



Warning

If you re-deploy the entire image, keep in mind that you need to have your data directory (`/srv/obs`) on a separate storage (LVM volume, partition etc.) otherwise it will be deleted!

1.3 Back-end Administration

1.3.1 Services

You can control the different back-end components via **`systemctl`**. You can enable/disable the service during booting the system and start/stop/restart it in a running system. For more information, see man page (https://www.freedesktop.org/software/systemd/man/systemctl.html#Commands). For example, to restart the repository server, use:

```
systemctl restart obsrepserver.service
```

TABLE 1.2: SERVICE NAMES

Component	Service Name
Source Server	obssrcserver.service
Repository Server	obsrepserver.service

Component	Service Name
Scheduler	obsscheduler.service
Dispatcher	obsdispatcher.service
Publisher	obspublisher.service
Source Publisher	obssourcepublish.service
Worker	obsworker.service
Source Services	obsservice.service
Download On Demand Repository Meta Data Updates	obsdodup.service
Delta Storage	obsdeltastore.service
Signer	obssigner.service
Warden	obswarden.service

1.3.2 Advanced Setups

It makes sense to run some of the different components of the OBS back-end on isolated hosts.

1.3.2.1 Distributed Workers

OBS workers can be very resource hungry. It all depends on the software that is being built, and how. Single builds deep down in the dependency chain can also trigger a sea of jobs. It makes sense to split off workers from all the other services so they do not have to fight for the same operating system/hardware resources. Here is an example on how to setup a remote OBS worker.

1. Install the worker package called obs-worker
2. Configure the OBS repository server address in the file /etc/sysconfig/obs-server. Change the variable OBS_REPO_SERVERS to the host name of the machine on which the repository server is running: OBS_REPO_SERVERS="myreposerver.example:5252"
3. Start the worker

1.4 Front-end Administration

The Ruby on Rails application is run through the Apache web server with mod_passenger (https://www.phusionpassenger.com/). You can control it via **`systemctl`**

```
systemctl {start, stop, restart} apache2
```

1.4.1 Delayed Jobs

Another component of the OBS front-end are delayed jobs for asynchronously executing longer tasks in the background. You can control this service also via **`systemctl`**.

```
systemctl {start, stop, restart} obsapidelayed
```

1.4.2 Full Text Search

The full-text search for packages and projects is handled by Thinking Sphinx (http://freelancing-gods.com/thinking-sphinx/). The delayed job daemon will take care of starting this service. To control it after boot, use the **`rake`** tasks it provides.

```
rake ts:{start, stop, rebuild, index}
```

2 Installation and Configuration

2.1 Planning

For testing your own OBS instance, or for small setups, such as if you only want to package a few scripts into RPMS and create proper installation sources from them, the ready-to-use obs-server appliance images are the easiest way. You can download them from http://openbuildservice.org/download/.

However, to use the OBS for large Linux software development with many packages, projects and users, consider setting up a regular installation. Depending on the number of users, projects, and architectures, you can split up the back-end (called partitioning) and have separate hosts for the front-end and the database.

For most installations, it is OK to run everything except workers on one host, if it has sufficient resources.

For flexibility and if you want some kind of high availability it is recommended to use virtualization for the different components.

2.1.1 Resource Planning

Normally, for an small or middle-sized installation, a setup with everything on one host (except workers) is sufficient. You should have a separate /srv volume for the back-end data. We recommend that you use XFS as file system.

For each scheduler architecture, you should add 4 GB RAM and one CPU core. For each build distribution you should add at least 50GB disk space per architecture.

A medium instance with about 50 users can easily run on a machine with 16GB RAM, 4 cores and 1 TB storage. The storage, of course, depends on the size of your projects and how often you have new versions.

For bigger installations, you can use separate networks for back-end communication, workers and front-end.

As of May 2021, the reference installation on **build.opensuse.org**, which has a lot of users and distributions, runs on a partitioned setup with:

- a mysql cluster as database
- api-server: 16GB RAM, 4 cores, 50GB disk
- separate binary back-ends (scheduler, dispatcher, reposerver, publisher, warden)
- source server: 11 GB RAM, 4 cores, 3 TB disk. The RAM is used mainly for caching.

- main back-end: 62 GB RAM (oversized), 16TB disk
- a lot of workers (see - https://build.opensuse.org/monitor)

For build time and performance, the count and performance of available worker hosts is more important than the other parts.

2.2 Simple Installation

In this document, we call "simple installation" an OBS installation where all OBS services are running on the same machine.

Important

It is very important that you read the **README.SETUP** file coming with your OBS version and follow the instructions there, because it may provide additional, version-specific information.

Before you start the installation of the OBS, you should make sure that your hosts have the correct fully qualified hostname, and that DNS is working and can resolve all names.

2.2.1 Back-end Installation

The back-end hosts all sources and built packages. It also schedules the jobs. To install it, install the "obs-server" package. After installation, it's a good idea to check the service configuration in **/usr/lib/obs/server/BSConfig.pm**, although the defaults should be good enough for simple cases.

Note

Read more about configuring the backend in *Section 2.4, “Distributed Setup”*.

The back-end consists of a number of systemd units (services):

TABLE 2.1: SERVICES

Service	Description	Remark
obssrcserver.service	Source server	
obsrepserver.service	Repository server	

Service	Description	Remark
obsservice.service	Source services server	
obsdodup.service	Repository metadata download	since 2.7
obsdeltastore.service	Delta storage	since 2.7
obsscheduler.service	Scheduler	
obsdispatcher.service	Dispatcher proxy	
obsservicedis-patch.service	Dispatcher	
obspublisher.service	Publisher	
obssigner.service	Signer proxy	
obssignd.service	Signer	
obswarden.service	Warden	
obscloduploadwork-er.service	Cloud upload worker	Only needed for cloud up-load feature
obscloduploadserv-er.service	Cloud upload server	Only needed for cloud up-load feature

These services are controlled via *systemctl*. Basically, you can enable/disable a service to start when the system boot, and you can start/stop/restart it in a running system as well. For more information, see the systemctl man page (https://manpages.opensuse.org/Tumbleweed/systemd/systemctl.1.en.html#COMMANDS). For example, to restart the repository server, do:

```
systemctl restart obsrepserver.service
```

When starting the various services, *obssrcserver.service* (the source server) must be started first, and *obsrepservice.service* (the repository server) second, followed by the remaining services in any order. When installing manually, you will need to first enable the services with

```
systemctl enable <name>
```

so they start automatically at boot. In this case, the start order will be enforced via the respective systemd unit files. Should you want to start the services manually, you will need to ensure the correct ordering yourself, by starting the source server first and the repository server second, like so:

```
systemctl start obssrcserver.service
systemctl start obsrepserver.service
```

followed by the remaining services in any order.



Warning

The start-up commands start services which are accessible from the outside. If the system is connected to an untrusted network, either block the ports with a firewall or do not run the commands at all.

2.2.1.1 Cloud Upload Setup

In order to setup the Cloud Upload feature you will need to configure the tools required per each cloud provider. Right now we only support the AWS Amazon Cloud (https://aws.amazon.com) and Microsoft Azure (https://portal.azure.com) as providers.

Before you can start uploading images to the Amazon Web Services (AWS) and/or Microsoft Azure, you have to:

1. Install the obs-cloud-uploader package

   ```
   zypper in obs-cloud-uploader
   ```

2. Start the cloud upload services

   ```
   systemctl start obscloduploadworker.service
   systemctl start obscloduploadserver.service
   ```

At last you have to register the cloud uploader service in */usr/lib/obs/server/BSConfig.pm*, for example, by adding following line:

```
our $cloduploadserver = "http://$hostname:5452";
```



Warning

Ensure that the system time of your cloud uploader instance is correct. AWS is relying on the timestamps of the requests it receives. Having an incorrect system time will cause cloud uploads to fail.

2.2.1.1.1 AWS Amazon Cloud

2.2.1.1.1.1 Authentication Workflow

We are going to use the role based authentication provided by Amazon to enable the OBS instance to upload images to other user's accounts.

The users will obtain an external ID (automatically created and unique) and the OBS account ID to create an Identity and Access Management (IAM) role. After the user created the role, he needs to provide the Amazon Resource Name (ARN) of the role to OBS. OBS will use this ARN to obtain temporary credentials, therefore an uploader account is necessary which we need to configure (see AWS authentication credentials setup). OBS will use the ARN to obtain temporary credentials for the users account to upload the appliance. The ARN and the external ID are not considered as a secret.

The whole workflow is described in the AWS documentation (https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_create_for-user_externalid.html).

2.2.1.1.1.2 Credentials Setup

For uploading images to AWS, OBS is using the AWS CLI (https://aws.amazon.com/cli) tool. Before you can start uploading your images, you have to enter the AWS credentials to the **/etc/obs/cloudupload/.aws/credentials** configuration file. These credentials will then be used by OBS to retrieve the temporary credentials from the ARN provided by users. More information about IAM role base authorization can be found in the Amazon documentation (https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_create_for-user_externalid.html).

2.2.1.1.2 Microsoft Azure

2.2.1.1.2.1 Authentication Workflow

The authentication is done via Microsoft's Active Directory. The user has to create a new application and needs to provide those two credentials to OBS:

1. Application ID
   The Application ID is a unique ID that represents an Active Directory Application.
2. Application Key
   The Application Key can be generated for every application and is the password.

OBS communicates with the REST API of Microsoft Azure to authenticate and upload images.

2.2.1.1.2.2 Configuration

The **Application ID** and the **Application Key** will be stored encrypted in the database. As for that, it's required to generate an SSL secret and public key that has to be stored on the server where the **obs-cloud-uploader** package has been installed.

To generate that SSL certificate, execute the following commands:

```
cd /etc/obs/cloudupload
openssl genrsa -out secret.pem
openssl rsa -in secret.pem -out _pubkey -outform PEM -pubout
```

2.2.1.1.2.3 Credentials setup

It's important that the public key is named **_pubkey** and the secret key is named **secret.pem** and are kept in **/etc/obs/cloudupload**.

2.2.2 Front-end Installation

You need to install the "obs-api" package for this and a MySQL server.

2.2.2.1 MySQL Setup

Make sure that the mysql server is started on every system reboot (use "insserv mysql" for permanent start). You should run mysql_secure_installation and follow the instructions.

Create the empty production databases:

```
# mysql -u root -p
mysql> create database api_production;
mysql> quit
```

Use a separate MySQL user (for example, `obs`) for the OBS access:

```
# mysql -u root -p
mysql> create user 'obs'@'%' identified by 'TopSecretPassword';
mysql> create user 'obs'@'localhost' identified by 'TopSecretPassword';
mysql> GRANT all privileges ON api_production.*
      TO 'obs'@'%', 'obs'@'localhost';
mysql> FLUSH PRIVILEGES;
mysql> quit
```

Configure your MySQL user and password in the "production" section of the api config: **/srv/www/obs/api/config/database.yml**

Example:

```
# MySQL (default setup).  Versions 4.1 and 5.0 are recommended.
#
# Get the fast C bindings:
#   gem install mysql
#   (on OS X: gem install mysql -- --include=/usr/local/lib)
# And be sure to use new-style password hashing:
#   http://dev.mysql.com/doc/refman/5.0/en/old-client.html

production:
  adapter: mysql2
  database: api_production
  username: obs
  password: TopSecretPassword
  encoding: utf8
  timeout: 15
  pool: 30
```

Now populate the database

```
cd /srv/www/obs/api/
sudo RAILS_ENV="production" rake db:setup
```

```
sudo RAILS_ENV="production" rake writeconfiguration
sudo chown -R wwwrun.www log tmp
```

Now you are done with the database setup.

2.2.2.2 Apache Setup

Now we need to configure the Web server. By default, you can reach the familiar web user interface and also api both on port 443 speaking https. Repositories can be accessed via http on port 82 (once some packages are built). An overview page about your OBS instance can be found behind *'http://localhost'*.

The obs-api package comes with an Apache vhost file, which does not need to get modified when you stay with these defaults: */etc/apache2/vhosts.d/obs.conf*

Install the required packages via

```
zypper in obs-api apache2 apache2-mod_xforward rubygem-passenger-apache2 memcached
```

Add the following Apache modules in `/etc/sysconfig/apache2`:

```
APACHE_MODULES="... passenger rewrite proxy proxy_http xforward headers socache_shmcb"
```

Enable SSL in /etc/sysconfig/apache2 via

```
APACHE_SERVER_FLAGS="SSL"
```

For production systems you should order official SSL certificates. For testing follow the instructions to create a self signed SSL certificate:

```
mkdir /srv/obs/certs
openssl genrsa -out /srv/obs/certs/server.key 4096
openssl req -new -key /srv/obs/certs/server.key \
        -out /srv/obs/certs/server.csr
openssl x509 -req -days 365 -in /srv/obs/certs/server.csr \
        -signkey /srv/obs/certs/server.key -out /srv/obs/certs/server.crt
cat /srv/obs/certs/server.key /srv/obs/certs/server.crt \
      > /srv/obs/certs/server.pem
```

At the time of this writing (2024), we consider the 4K RSA key to be a safe implementation, but you might want to check out the current standards by consulting (on both client and server side) the output of

```
man crypto-policies
```

and

```
man update-crypto-policies
```

To allow the usage of https API in Web UI code you need to trust your certificate as well:

```
cp /srv/obs/certs/server.pem /etc/ssl/certs/
c_rehash /etc/ssl/certs/
```

2.2.2.3 API Configuration

Check and edit **/srv/www/obs/api/config/options.yml**

If you change the hostnames/ips of the API, you need to adjust **frontend_host** accordingly. If you want to use LDAP, you need to change the LDAP settings as well. Look at the *Section 5.8, "Managing Users and Groups"* for details. You will find examples and more details in the *Section 3.1, "Configuration Files"*.

It is strongly recommended to enable

```
use_xforward: true
```

as well here, to tell Rails to forward requests to the back-end for asynchronous processing. (Without this setting, the front-end will block while the back-end handles each request.)

Afterwards, you can start the OBS API and make it permanent via

```
systemctl enable apache2
systemctl start apache2

systemctl enable obs-api-support.target
systemctl start obs-api-support.target

systemctl enable memcached.service
systemctl start memcached.service
```

Now you have you own empty instance running and you can do some online configuration steps.

2.2.3 Online Configuration

To customize the OBS instance you may need to configure some settings via the OBS API and Web user interface.

First you should change the password of the Admin account, for this you need first login as user Admin in the Web UI with the default password "opensuse". Click on the Admin link (right top of the page), here you can change the password.

After changing the Admin password, set up **osc** to use the Admin account for more changes. Here an example:

```
osc -c ~/.obsadmin_osc.rc -A https://api.testobs.org
```

Follow the instructions on the terminal.



Warning

The password is stored in clear text in this file by default, so you need to give this file restrictive access rights, only read/write access for your user should be allowed. **osc** allows to store the password in other ways (in keyrings for example), refer to the osc documentation for this.

Now you can check out the main configuration of the OBS:

```
osc -c ~/.obsadmin_osc.rc api /configuration >/tmp/obs.config
cat /tmp/obs.config
<configuration>
  <title>Open Build Service</title>
  <description>
    &lt;p class="description"&gt;
      The &lt;a href="http://openbuildservice.org"&gt; Open Build Service (OBS)&lt;/a&gt;
      is an open and complete distribution development platform that provides a
 transparent
      infrastructure for development of Linux distributions, used by openSUSE, MeeGo
      and other distributions.
      Supporting also Fedora, Debian, Ubuntu, RedHat and other Linux distributions.
    &lt;/p&gt;
    &lt;p class="description"&gt;
      The OBS is developed under the umbrella of the &lt;a href="http://
www.opensuse.org"&gt;openSUSE project&lt;
      /a&gt;. Please find further informations on the &lt;
      a href="http://wiki.opensuse.org/openSUSE:Build_Service"&gt;openSUSE Project wiki
 pages&lt;/a&gt;.
    &lt;/p&gt;

    &lt;p class="description"&gt;
      The Open Build Service developer team is greeting you. In case you use your OBS
 productive
      in your facility, please do us a favor and add yourself at &lt;
      a href="http://wiki.opensuse.org/openSUSE:Build_Service_installations"&gt;
      this wiki page&lt;/a&gt;. Have fun and fast build times!
    &lt;/p&gt;
  </description>
```

```
  <name>private</name>
  <download_on_demand>on</download_on_demand>
  <enforce_project_keys>off</enforce_project_keys>
  <anonymous>on</anonymous>
  <registration>allow</registration>
  <default_access_disabled>off</default_access_disabled>
  <allow_user_to_create_home_project>on</allow_user_to_create_home_project>
  <disallow_group_creation>off</disallow_group_creation>
  <change_password>on</change_password>
  <hide_private_options>off</hide_private_options>
  <gravatar>on</gravatar>
  <cleanup_empty_projects>on</cleanup_empty_projects>
  <disable_publish_for_branches>on</disable_publish_for_branches>
  <admin_email>unconfigured@openbuildservice.org</admin_email>
  <unlisted_projects_filter>^home:.+</unlisted_projects_filter>
  <unlisted_projects_filter_description>home projects</
unlisted_projects_filter_description>
  <schedulers>
    <arch>armv7l</arch>
    <arch>i586</arch>
    <arch>x86_64</arch>
  </schedulers>
</configuration>
```



Important

unlisted_projects_filter only admit Regular Expression (see RLIKE specifications of MySQL/MariaDB for more information) and **unlisted_projects_filter_description** is part of the link shown in the project list for filtering

You should edit this file according to your preferences, then sent it back to the server:

```
osc -c ~/.obsadmin_osc.rc api /configuration -T /tmp/obs.config
```

If you want to use an interconnect to another OBS instance to reuse the build targets you can do this as Admin via the Web UI or create a project with a **remoteurl** tag (see *Section 3.4.2, "Project Metadata"*)

```
<project name="openSUSE.org">
  <title>openSUSE.org Project</title>
  <description>
 This project refers to projects hosted on the Build Service
[...]

Use openSUSE.org:openSUSE:12.3 for example to build against the
```

```
openSUSE:12.3 project as specified on the opensuse.org Build Service.
</description>
  <remoteurl>https://api.opensuse.org/public</remoteurl>
</project>
```

You can create the project using a file with the above content with **osc** like this:

```
osc -c ~/.obsadmin_osc.rc meta prj openSUSE.org -F /tmp/openSUSE.org.meta
```

You also can import binary distribution, see *Section 5.2.2, "Importing Distributions"* for this.

The OBS has a list of available distributions used for build. This list is displayed to user, if they are adding repositories to their projects. This list can be managed via the API path /distributions

```
osc -c ~/.obsadmin_osc.rc api /distributions > /tmp/distributions.xml
```

Example distributions.xml file:

```
<distributions>
  <distribution vendor="SUSE" version="SLE-12-SP1" id="137">
    <name>SLE-12-SP1</name>
    <project>SUSE:SLE-12-SP1</project>
    <reponame>SLE-12-SP1</reponame>
    <repository>standard</repository>
    <link>http://www.suse.com/</link>
    <icon url="https://static.opensuse.org/distributions/logos/suse-SLE-12-8.png"
 width="8" height="8"/>
    <icon url="https://static.opensuse.org/distributions/logos/suse-SLE-12-16.png"
 width="16" height="16"/>
    <architecture>x86_64</architecture>
  </distribution>
</distributions>
```

You can add your own distributions here and update the list on the server:

```
osc -c ~/.obsadmin_osc.rc api /distributions -T /tmp/distributions.xml
```

2.3 Worker Farm

To not burden your OBS back-end daemons with the unpredictable load package builds can produce (think someone builds a monstrous package like LibreOffice) you should not run OBS workers on the same host as the rest of the back-end daemons.



Important

You back-end need to be configured to use the correct hostnames for the repo and source server and the ports need to be reachable by the workers. Also, the IP addresses of the workers need to be allowed to connect the services. (look at the */usr/lib/obs/server/BSConfig.pm::ipaccess* array).

You can deploy workers quite simply using the worker appliance. Or install a minimum system plus the obs-worker package on the hardware.

Edit the **/etc/sysconfig/obs-server** file, at least **OBS_SRC_SERVER**, **OBS_REPO_SERVERS** and **OBS_WORKER_INSTANCES** need to be set. More details in the *Section 3.1, "Configuration Files"*.

start the worker:

```
systemctl enable obsworker
systemctl start obsworker
```

2.4 Distributed Setup

All OBS back-end daemons can also be started on individual machines in your network. Also, the front-end Web server and the MySQL server can run on different machines. Especially for large scale OBS installations this is the recommended setup.

A setup with partitioning is very similar to the steps of the simple setup. Here we are only mention the differences to the simple setup.



Note

You need to make sure that the different machines can communicate via the network, it is very recommended to use a separate network for this to isolate it from the public part.

On all back-end hosts you need to install the obs-server package. On the front-end host you need to install the obs-api package.



Important

Only one source server instance can be exist on a single OBS installation.

The binary back-end can be split on project level, this is called partitioning.

On one partition following services needs to be configured and run:

1. repserver
2. schedulers
3. dispatcher
4. warden
5. publisher

You do not need to share any directories on File System level between the partitions.

Here some example for partitioning:

1. A main partition for everything not in the others (host mainbackend)
2. A home partition for all home projects of the users (host homebackend)
3. A release partition for released software projects (host releasebackend)

The configuration is done in the back-end config file **/usr/lib/obs/server/BSConfig.pm**. Most parts of the file can be shared between the back-ends.

Here the important parts of the mainbackend of out testobs.org installation:

```
[...]
my $hostname = Net::Domain::hostfqdn() || 'localhost';
# IP corresponding to hostname (only used for $ipaccess); fallback to localhost since
 inet_aton may fail to resolve at shutdown.
my $ip = quotemeta inet_ntoa(inet_aton($hostname) || inet_aton("localhost"));

my $frontend = 'api.testobs.org'; # FQDN of the Web UI/API server if it's not $hostname

# If defined, restrict access to the backend servers (bs_repserver, bs_srcserver,
 bs_service)
our $ipaccess = {
   '127\..*' => 'rw', # only the localhost can write to the backend
   "^$ip" => 'rw',    # Permit IP of FQDN
   "10.20.1.100" => 'rw',    # Permit IP of srcsrv.testobs.org
   "10.20.1.101" => 'rw',    # Permit IP of mainbackend.testobs.org
   "10.20.1.102" => 'rw',    # Permit IP of homebackend.testobs.org
   "10.20.1.103" => 'rw',    # Permit IP of releasebackend.testobs.org
   '10.20.2.*' => 'worker',  # build results can be delivered from any client in the
 network
};
```

```
# IP of the Web UI/API Server (only used for $ipaccess)
if ($frontend) {
  my $frontendip = quotemeta inet_ntoa(inet_aton($frontend) || inet_aton("localhost"));
  $ipaccess->{$frontendip} = 'rw' ; # in dotted.quad format
}

# also change the SLP reg files in /etc/slp.reg.d/ when you touch hostname or port
our $srcserver = "http://srcsrv.testobs.org:5352";
our $reposerver = "http://mainbackend.testobs.org:5252";
our $serviceserver = "http://service.testobs.org:5152";

# Needed if you want to use the cloud upload feature
our $clouduploadserver = "http://$hostname:5452";

#
our @reposervers = ("
    http://mainbackend.testobs.org:5252,
    http://homebackend.testobs.org:5252,
    http://releasebackend.testobs.org:5252
");

# you can use different ports for worker connections
our $workersrcserver = "http://w-srcsrv.testobs.org:5353";
our $workerreposerver = "http://w-mainbackend.testobs.org:5253";
[...]
our $partition = 'main';
#
# this defines how the projects are split. All home: projects are hosted
# on an own server in this example. Order is important.
our $partitioning = [
    'home:' => 'home',
    'release' => 'release'
    '.*'    => 'main',
];
our $partitionservers = {
    'home' => 'http://homebackend.testobs.org:5252',
    'release' => 'http://releasebackend.testobs.org:5252',
    'main' => 'http://mainbackend.testobs.org:5252',
};
[...]
```

On the other partition server you need to change "**our $reposerver**", "**our $workerreposerver**" and "**our $partition**".

On all partition servers you need to start:

```
systemctl start obsrepserver.service
```

```
systemctl start obsscheduler.service
systemctl start obsdispatcher.service
systemctl start obspublisher.service
systemctl start obswarden.service
```

On the worker machines you should set of repo servers in the **OBS_REPO_SERVERS** variable. You can also define workers with a subset of the repo servers to prioritize partitions.

2.5 Monitoring

In this chapter you will find some general monitoring instructions for the Open Build Service. All examples are based on Nagios plugins, but the information provided should be easily adaptable for other monitoring solutions.

2.5.1 Endpoint Checks

2.5.1.1 HTTP Checks: Checking Whether the HTTP Server Responds

This check will output a critical if the HTTP server with ip address 172.19.19.19 (-I 172.19.19.19) listening on port 80 (-p 80) does not answer and output a warning if the HTTP return code is not 200. The server name that will be used is server (-H server) which is important if different virtual hosts are listening on the same port.

```
check_http -H server -I 172.19.19.19 -p 80 -u http://server
```

The same check, but this time it will check a ssl enabled HTTP server.

```
check_http -S -H server -I 172.19.19.19 -p 443 -u https://server
```

It is also possible to check the presence of a certain string in the HTTP response. In this case it will check for the string *Source Service Server*.

```
check_http -s "Source Service Server" -S -H server -I 172.19.19.19 -p 5152
```

Open Build Service HTTP endpoints that should be checked:

1. Web Interface / API: port 443
2. Repository Server: port 82

3. Package Repository Server: port 5252
4. Source Repository Server: port 5352
5. Source Service Server: port 5152
6. Cloud Upload Server: port 5452

2.5.2 Common Checks

This is a list of common checks that should be run on each individual server.

2.5.2.1 Disk Space: Checking Available Disk Space

This check will output a warning if less than 10 percent disk space is available (-w 10) and output a critical if less than 5 percent disk space are available (-c 5). It will check all file systems except file systems with type *none* (-x none).

```
check_disk -w 10 -c 5 -x none
```

2.5.2.2 Memory Usage: Checking Available Memory

This check will output a warning if less than 10 percent memory is available (-w 10) and output a critical if less than 5 percent memory is available (-c 5). OS caches will be counted as free memory (-C) and it will check the available memory (-f). check_mem.pl is not a standard Nagios plugin and can be downloaded at https://exchange.nagios.org/.

```
check_mem.pl -f -C -w 10 -c 5
```

2.5.2.3 NTP: Checking Date and Time

This check will compare the local time with the time provided by the NTP server pool.ntp.org (-H pool.ntp.org). It will output a warning if the time differs by 0.5 seconds (-w 0.5) and output a critical if the time differs by 1 seconds (-c 1).

```
check_ntp_time -H pool.ntp.org -w 0.5 -c 1
```

2.5.2.4 Ping: Checking That the Server Is Alive

This plugin checks if the server responds to a ping request and it will output a warning if the respond time exceeds 200ms or 30 percent package loss (-w 200.0,30%) and output a critical if the respond time exceeds 500ms or 60 percent package loss.

```
check_icmp -H server -w 200.0,30% -c 500.0,60%
```

2.5.2.5 Load: Checking the Load on the Server

This check will output a warning if the load value exceeded 7.0 in the last minute, 6.0 in the last 5 minutes or 5.0 in the last 15 minutes (-w 7.0,6.0,5.0). It will output a critical if the load value exceeded 12.0 in the last minute, 8.0 in the last 5 minutes or 6.0 in the last 15 minutes (-c 12.0,8.0,6.0).

```
check_load -w 7.0,6.0,5.0 -c 12.0,8.0,6.0
```

2.5.2.6 Disk Health: Checking the Health of Local Hard Disks

This check is only relevant on physical systems with local storage attached to it. It will check the disk status utilizing the S.M.A.R.T interface and it will output a critical if any of the S.M.A.R.T values exceeds critical limits. check_smartmon is not a standard Nagios plugin and can be downloaded at https://exchange.nagios.org/.

```
check_smartmon --drive /dev/sda --drive /dev/sdb
```

2.5.3 Other Checks

2.5.3.1 MySQL: Checking That the MySQL Database Is Responding

This check will check that the MySQL database server is running and that the database *api_production* is available.

```
check_mysql -H localhost -u nagios -p xxxxxx -d api_production
```

MySQL Databases to check:

1. api_production
2. mysql

2.5.3.2 Backup Status: Checking That a Valid Backup Is Available

It is always advisable to check that the last backup run was successful and a recent backup is available. The check itself depends on the Backup solution that is used.

3 File System Overview

3.1 Configuration Files

3.1.1 Front-end Configuration

The front-end is configured with four files:

- /srv/www/obs/api/config/database.yml
- /srv/www/obs/api/config/options.yml
- /srv/www/obs/api/config/feature.yml
- /etc/apache2/vhosts.d/obs.conf

3.1.1.1 `database.yml`

This file has the information needed to access the database. It contain credentials for the database access and should be only readable by root and the group running the Web server (www).

The file has settings for the production, development and test ruby environment, for production systems only the production section is important.

Example production section

```
production:
  adapter: mysql2
  database: api_production
  username: obsapiuser
  password: topsecret
  encoding: utf8
  timeout: 15
  pool: 30
```

TABLE 3.1: DATABASE CONFIGURATION KEYWORDS

keyword	Description	Remarks
adapter	Database driver	only MySQL databases are supported

keyword	Description	Remarks
database	Database name	do not change !
username	MySQL user name	database user, not a system user
password	password for this user	clear text
encoding	codetable	
timeout	wait time in milliseconds	
pool	number of open connections per thread	
socket	path to the MySQL socket	same host only
host	IP address or hostname of the MySQL server	for remote servers
port	port number of the MySQL server	for remote servers

3.1.1.2 `options.yml`

The configuration file /srv/www/obs/api/config/options.yml is the default configuration file for the Open Build Service Web UI and API. It contains configuration parameters for example for back-end connections and connection to the API. Important are the configurations for source and front-end hosts. The configuration for LDAP authentication is also located in this file.

Configuration options can be set per Rails environment (https://guides.rubyonrails.org/configuring.html#rails-environment-settings) or as generic configuration option defined in *default*.



Note

We've updated the format of the *options.yml* after the release of OBS 2.9. Old configuration files can be converted via

```
(cd /srv/www/obs/api/; rake migrate_options_yml)
```



Note

More and more configurations will be moved to the database and do not longer exist in this file. The database configuration can be accessed via the API */configuration* path.

TABLE 3.2: `options.yml` CONFIGURATION ITEMS

Config item	Description	Values `default`	Remarks
use_xforward	Use mod_xforward module	`true` false	Apache only, should be true
use_nginx_redirect	Use X-Accel-Redirect	`/inter-nal_redirect`	Nginx only
min_votes_for_rating	Minimum votes for a rating	integer `3`	
response_schema_valida-tion	Set to true to verify XML responses comply to the schema	true `false`	test/debug option
source_host	back-end source server host	`localhost`	
source_port	back-end source server port	integer `5352`	
source_protocol	back-end source server protocol	`http`, https	
front end_host	Front-end host	`localhost`	
frontend_port	Front-end port	integer `443`	
frontend_protocol	Front-end protocol	http `https`	
external_frontend_host	External Front-end host		if your users access the hosts through a proxy or different name

Config item	Description	Values `default`	Remarks
external_frontend_port	External Front-end port	integer `443`	
external_frontend_protocol	External Front-end protocol	http `https`	
extended_backend_log	Extended back-end log	`true` false	test/debug option
proxy_auth_mode:	turn proxy mode on/off	`:off` :on	see LDAP section
proxy_auth_test_user	Test user	`coolguy`	test/debug option
proxy_auth_test_email	Email of Test user	`coolguy@ example.com`	test/debug option
global_write_through	if set to false, the API will only fake writes to back-end	`true` false	test/debug option
auto_cleanup_after_days	not longer used	`30`	moved to /configuration API
errbit_api_key	API key of the application		test/debug option
errbit_host	installation of errbit.com a Ruby error catcher		test/debug option
errbit_api_key	API key of the application		test/debug option
ldap_mode:	OBS LDAP mode on/off	`:off` :on	see LDAP section

Example options.yml

```
#
# This file contains the default configuration of the Open Build Service
# API.
```

```
#

default: &default
  # Make use of mod_xforward module in apache
  use_xforward: true

  # Make use of X-Accel-Redirect for Nginx.
  # http://kovyrin.net/2010/07/24/nginx-fu-x-accel-redirect-remote
  #use_nginx_redirect: /internal_redirect

  # Minimum count of rating votes a project/package needs to # be taken in
  # account
  # for global statistics:
  min_votes_for_rating: 3

  # Set to true to verify XML reponses comply to the schema
  response_schema_validation: false

  # backend source server
  source_host: localhost
  source_port: 5352
  #source_protocol: https

  # api access to this instance
  frontend_host: localhost
  frontend_port: 443
  frontend_protocol: https
  # if your users access the hosts through a proxy (or just a different name,
  # use this to
  # overwrite the settings for users)
  #external_frontend_host: api.opensuse.org
  #external_frontend_port: 443
  #external_frontend_protocol: https


  extended_backend_log: true

  # proxy_auth_mode can be :off, :on or :simulate
  proxy_auth_mode: :off

  # ATTENTION: If proxy_auth_mode'is :on, the frontend takes the user
  # name that is coming as headervalue X-username as a
  # valid user does no further authentication. So take care...
  proxy_auth_test_user: coolguy
  proxy_auth_test_email: coolguy@example.com

  # set this to enable auto cleanup requests after the given days
```

```
  auto_cleanup_after_days: 30

  #schema_location

  #version

  # if set to false, the API will only fake writes to backend (useful in
  # testing)
  # global_write_through: true

  # see
  # http://colszowka.heroku.com/2011/02/22/setting-up-your-custom-hoptoad-notifier-
endpoint-for-free-using-errbit-on-heroku
  #errbit_api_key: api_key_of_your_app
  #errbit_host: installation.of.errbit.com

production:
  <<: *default

test:
  <<: *default
  source_host: backend
  memcached_host: cache

development:
  <<: *default
  source_host: backend
  memcached_host: cache
```

3.1.1.3 `feature.yml`

The configuration file /srv/www/obs/api/config/feature.yml contains the default configuration about features that can be enabled or disabled in Open Build Service.

TABLE 3.3: `feature.yml` CONFIGURATION ITEMS

Config item	Description	Values `default`	Remarks
image_templates	enable/disable image template feature	`true` false	see Reference Guide for more information
kiwi_image_editor	enable/disable Kiwi Image Editor	true `false`	

Config item	Description	Values `default`	Remarks
cloud_upload	enable/disable Cloud Upload setup	true `false`	

Example feature.yml

```
production:
  features: &default
    image_templates: true
    kiwi_image_editor: false
    cloud_upload: false

development:
  features:
    <<: *default
    kiwi_image_editor: true
    cloud_upload: true

test:
  features:
    <<: *default
    kiwi_image_editor: true
    cloud_upload: true
```

3.1.1.4 Apache Virtual Host `obs.conf`

The Apache configuration depends on the Apache version and which extra options are used, so use the documentation of the Apache version you are using.

Here, as an example, the standard configuration used by the appliance: Apache vhost example

```
Listen 82
# May needed on old distributions or after an update from them.
#Listen 443

# Passenger defaults
PassengerSpawnMethod "smart"
PassengerMaxPoolSize 20
#RailsEnv "development"

# allow long request urls and being part of headers
LimitRequestLine 20000
LimitRequestFieldsize 20000
```

```
# Just the overview page
<VirtualHost *:80>
        # just give an overview about this OBS instance via static web page
        DocumentRoot  "/srv/www/obs/overview"

        <Directory /srv/www/obs/overview>
           Options Indexes
           Require all granted
        </Directory>
</VirtualHost>


# Build Results
<VirtualHost *:82>
        # The resulting repositories
        DocumentRoot  "/srv/obs/repos"

        <Directory /srv/obs/repos>
           Options Indexes FollowSymLinks
           Require all granted
        </Directory>
</VirtualHost>


# OBS WEB UI & API
<VirtualHost *:443>
        ServerName api

        #  General setup for the virtual host
        DocumentRoot  "/srv/www/obs/api/public"
        ErrorLog /srv/www/obs/api/log/apache_error.log
        TransferLog /srv/www/obs/api/log/apache_access.log

        PassengerMinInstances 2
        PassengerPreStart https://api

        SSLEngine on

        #  SSL protocols
        #  Supporting TLS only is adequate nowadays
        SSLProtocol all -SSLv2 -SSLv3

        #   SSL Cipher Suite:
        #   List the ciphers that the client is permitted to negotiate.
        #   We disable weak ciphers by default.
        #   See the mod_ssl documentation or "openssl ciphers -v" for a
        #   complete list.
```

```
        SSLCipherSuite ALL:!aNULL:!eNULL:!SSLv2:!LOW:!EXP:!MD5:@STRENGTH

        SSLCertificateFile /srv/obs/certs/server.crt
        SSLCertificateKeyFile /srv/obs/certs/server.key

        <Directory /srv/www/obs/api/public>
           AllowOverride all
           Options -MultiViews

           # This requires mod_xforward loaded in apache
           # Enable the usage via options.yml
           # This will decrease the load due to long running requests a lot (unloading
 from rails stack)
           XForward on

           Require all granted
        </Directory>

        SetEnvIf User-Agent ".*MSIE [1-5].*" \
                 nokeepalive ssl-unclean-shutdown \
                 downgrade-1.0 force-response-1.0

        CustomLog /var/log/apache2/ssl_request_log   ssl_combined


        # from http://guides.rubyonrails.org/asset_pipeline.html
        <LocationMatch "^/assets/.*$">
          Header unset ETag
          FileETag None
          # RFC says only cache for 1 year
          ExpiresActive On
          ExpiresDefault "access plus 1 year"
       </LocationMatch>

        SetEnvIf User-Agent ".*MSIE [1-5].*" \
                 nokeepalive ssl-unclean-shutdown \
                 downgrade-1.0 force-response-1.0

        ## Older firefox versions needs this, otherwise it wont cache anything over SSL.
        Header append Cache-Control "public"

</VirtualHost>
```

3.1.2 Back-end Configuration

The Back-end is configured with 2 files:

- /etc/sysconfig/obs-server - a shell script used for workers and the OBS start scripts
- /usr/lib/obs/server/BSConfig.pm - a Perl script defining some global variables

3.1.2.1 `/etc/sysconfig/obs-server`

This script is used to set up the basic paths and the worker. the most important settings are the **OBS_SRC_SERVER** and **OBS_REPO_SERVERS** and the **OBS_WORKER_INSTANCES**.

TABLE 3.4: `obs-server` VARIABLES

Variable	Description	Values `default`	Remarks
OBS_BACKENDCODE_DIR	Path to the back-end scripts	/usr/lib/obs/server/	
OBS_RUN_DIR	communication directory base	/srv/obs/run	
OBS_LOG_DIR	logging directory	/srv/obs/log	
OBS_BASE_DIR	base directory	/srv/obs	
OBS_API_AUTOSETUP	Automatically setup API and Web UI	yes `no`	appliance only, will overwrite config files
OBS_SRC_SERVER	source server host	localhost:5352	only one
OBS_REPO_SERVERS	repository server host	localhost:5252	maybe a list
OBS_WORKER_INSTANCES	number of build instances	integer `0`	
OBS_WORKER_INSTANCE_NAMES	names of the workers		space-separated list

Variable	Description	Values `default`	Remarks
OBS_WORKER_DIRECTO-RY	worker base directo-ry		
OBS_WORKER_PORTBASE	The base for port numbers used by worker	integer `0`	0 OS assign number
OBS_WORKER_JOBS	Number of parallel compile jobs	integer `1`	
OBS_WORK-ER_TEST_MODE	Run in test mode	yes `no`	
OBS_WORKER_HOST LA-BELS	one or more labels for the build host		may used by con-straints
OBS_USE_SLP	Register in SLP serv-er	`yes` no	
OBS_CACHE_DIR	cache directory for downloaded pack-ages		
OBS_CACHE_SIZE	package cache size		in MB
OBS_WORKER_NICE _LEV-EL	nice level of running workers	`18`	
OBS_VM_TYPE	VM type	**auto** Xen kvm lxc zvm emulator:$arch none	
OBS_VM_KERNEL	Set kernel used by worker	`none` (/boot/vmlin-uz)	KVM option
OBS_VM_INITRD	initrd used by worker	`none` (/boot/vmlin-uz)	KVM option

Variable	Description	Values `default`	Remarks
OBS_VM_DISK_AUTOSET-UP _ROOT_FILESIZE	Autosetup disk size	`4096`	in MB
OBS_VM_DISK_AUTOSET-UP _SWAP_FILESIZE	Autosetup swap size	`1024`	on MB
OBS_VM_DISK_AUTOSET-UP _FILESYSTEM	File System used with autosetup	`ext3`	
OBS_VM_DISK_AUTOSET-UP _MOUNT_OPTIONS	Special mount options		
OBS_VM_USE_TMPFS	Enable build in memory	yes `no`	requires much memory
OBS_INSTANCE_MEMORY	Memory allocated for a VM	`512`	
OBS_STORAGE_AUTOSET-UP	storage auto configuration	yes `no`	may destroy disk content
OBS_SETUP_WORKER _PARTITIONS	LVM via obsstorage-setup	take_all `use_obs_vg` none	may destroy disk content
OBS_WORK-ER_CACHE_SIZE	LVM partition for cache size		
OBS_WORKER_ROOT_SIZE	LVM partition for root size		
OBS_WORKER_SWAP_SIZE	LVM partition for swap size		
OBS_WORKER_BINARIES _PROXY	proxy service for caching binaries		
OBS_ROOT_SSHD_KEY_URL	ssh pub key to allow root user login		for mass deployment

Variable	Description	Values `default`	Remarks
OBS_WORKER_SCRIP-T_URL	URL to the initial script		

For workers the settings could be declared in the */etc/buildhost.config* file as well.

```
#
# NOTE: all these options can be also declared in /etc/buildhost.config on each worker
 differently.
#

## Path:        Applications/OBS
## Description: The OBS backend code directory
## Type:        string
## Default:     ""
## Config:      OBS
#
# An empty dir will lead to the fall back directory, typically /usr/lib/obs/server/
#
OBS_BACKENDCODE_DIR=""

## Path:        Applications/OBS
## Description: The base for OBS communication directory
## Type:        string
## Default:     ""
## Config:      OBS
#
# An empty dir will lead to the fall back directory, typically /srv/obs/run
#
OBS_RUN_DIR="/srv/obs/run"

## Path:        Applications/OBS
## Description: The base for OBS logging directory
## Type:        string
## Default:     ""
## Config:      OBS
#
# An empty dir will lead to the fall back directory, typically /srv/obs/log
#
OBS_LOG_DIR="/srv/obs/log"

## Path:        Applications/OBS
## Description: The base directory for OBS
## Type:        string
## Default:     ""
```

```
## Config:      OBS
#
# An empty dir will lead to the fall back directory, typically /srv/obs
#
OBS_BASE_DIR=""

## Path:        Applications/OBS
## Description: Automatically set up API and Web UI for OBS server, be warned, this will
 replace config files!
## Type:        ("yes" | "no")
## Default:     "no"
## Config:      OBS
#
# This is usually only enabled on the OBS Appliance
#
OBS_API_AUTOSETUP="yes"
#
# NOTE: all these options can be also declared in /etc/buildhost.config on each worker
 differently.
#

## Path:        Applications/OBS
## Description: define source server host to be used
## Type:        string
## Default:     ""
## Config:      OBS
#
# An empty setting will point to localhost:5352 by default
#
OBS_SRC_SERVER=""

## Path:        Applications/OBS
## Description: define repository server host to be used
## Type:        string
## Default:     ""
## Config:      OBS
#
# An empty setting will point to localhost:5252 by default
#
OBS_REPO_SERVERS=""

## Path:        Applications/OBS
## Description: define number of build instances
## Type:        integer
## Default:     0
## Config:      OBS
#
```

```
# 0 instances will automatically use the number of CPU's
#
OBS_WORKER_INSTANCES="0"

## Path:        Applications/OBS
## Description: define names of build instances for z/VM
## Type:        string
## Default:     ""
## Config:      OBS
#
# The names of the workers as defined in z/VM. These must have two minidisks
# assigned, and have a secondary console configured to the local machine:
# 0150 is the root device
# 0250 is the swap device
#
#OBS_WORKER_INSTANCE_NAMES="LINUX075 LINUX076 LINUX077"
OBS_WORKER_INSTANCE_NAMES=""

## Path:        Applications/OBS
## Description: The base directory, where sub directories for each worker will get
 created
## Type:        string
## Default:     ""
## Config:      OBS
#
#
OBS_WORKER_DIRECTORY=""

## Path:        Applications/OBS
## Description: The base for port numbers used by worker instances
## Type:        integer
## Default:     "0"
## Config:      OBS
#
# 0 means let the operating system assign a port number
#
OBS_WORKER_PORTBASE="0"

## Path:        Applications/OBS
## Description: Number of parallel compile jobs per worker
## Type:        integer
## Default:     "1"
## Config:      OBS
#
# this maps usually to "make -j1" during build
#
OBS_WORKER_JOBS="1"
```

```
## Path:        Applications/OBS
## Description: Run in test mode (build results will be ignore, no job blocking)
## Type:        ("yes" | "")
## Default:     ""
## Config:      OBS
#
OBS_WORKER_TEST_MODE=""

## Path:        Applications/OBS
## Description: define one or more labels for the build host.
## Type:        string
## Default:     ""
## Config:      OBS
#
# A label can be used to build specific packages only on dedicated hosts.
# For example for benchmarking.
#
OBS_WORKER_HOSTLABELS=""

## Path:        Applications/OBS
## Description: Register in SLP server
## Type:        ("yes" | "no")
## Default:     "yes"
## Config:      OBS
#
#
OBS_USE_SLP="yes"

## Path:        Applications/OBS
## Description: Use a common cache directory for downloaded packages
## Type:        string
## Default:     ""
## Config:      OBS
#
# Enable caching requires a given directory here. Be warned, content will be
# removed there !
#
OBS_CACHE_DIR=""

## Path:        Applications/OBS
## Description: Defines the package cache size
## Type:        size in MB
## Default:     ""
## Config:      OBS
#
# Set the size to 50% of the maximum usable size of this partition
```

```
#
OBS_CACHE_SIZE=""

## Path:        Applications/OBS
## Description: Defines the nice level of running workers
## Type:        integer
## Default:     18
## Config:      OBS
#
# Nicenesses range from -20 (most favorable  scheduling) to 19 (least
# favorable).
# Default to 18 as some testsuites depend on being able to switch to
# one priority below (19) _and_ having changed the numeric level
# (so going from 19->19 makes them fail).
#
OBS_WORKER_NICE_LEVEL=18

## Path:        Applications/OBS
## Description: Set used VM type by worker
## Type:        ("auto" | "xen" | "kvm" | "lxc" | "zvm" | "emulator:$arch" | "emulator:
$arch:$script" | "none")
## Default:     "auto"
## Config:      OBS
#
#
OBS_VM_TYPE="auto"

## Path:        Applications/OBS
## Description: Set kernel used by worker (kvm)
## Type:        ("none" | "/boot/vmlinuz" | "/foo/bar/vmlinuz)
## Default:     "none"
## Config:      OBS
#
# For z/VM this is normally /boot/image
#
OBS_VM_KERNEL="none"

## Path:        Applications/OBS
## Description: Set initrd used by worker (kvm)
## Type:        ("none" | "/boot/initrd" | "/foo/bar/initrd-foo)
## Default:     "none"
## Config:      OBS
#
# for KVM, you have to create with (example for openSUSE 11.2):
#
# export rootfstype="ext4"
```

```
# mkinitrd -d /dev/null -m "ext4 binfmt_misc virtio_pci virtio_blk" -k
 vmlinuz-2.6.31.12-0.2-default -i initrd-2.6.31.12-0.2-default-obs_worker
#
# a working initrd file which includes virtio and binfmt_misc for OBS in order to work
 fine
#
# for z/VM, the build script will create a initrd at the given location if
# it does not yet exist.
#
OBS_VM_INITRD="none"

## Path:        Applications/OBS
## Description: Autosetup for XEN/KVM/TMPFS disk (root) - Filesize in MB
## Type:        integer
## Default:     "4096"
## Config:      OBS
#
#
OBS_VM_DISK_AUTOSETUP_ROOT_FILESIZE="4096"

## Path:        Applications/OBS
## Description: Autosetup for XEN/KVM disk (swap) - Filesize in MB
## Type:        integer
## Default:     "1024"
## Config:      OBS
#
#
OBS_VM_DISK_AUTOSETUP_SWAP_FILESIZE="1024"

## Path:        Applications/OBS
## Description: Filesystem to use for autosetup {none,ext4}=ext4, ext3=ext3
## Type:        string
## Default:     "ext3"
## Config:      OBS
#
#
OBS_VM_DISK_AUTOSETUP_FILESYSTEM="ext3"

## Path:        Applications/OBS
## Description: Filesystem mount options to use for autosetup
## Type:        string
## Default:     ""
## Config:      OBS
#
#
OBS_VM_DISK_AUTOSETUP_MOUNT_OPTIONS=""
```

```
## Path:        Applications/OBS
## Description: Enable build in memory
## Type:        ("yes" | "")
## Default:     ""
## Config:      OBS
#
# WARNING: this requires much memory!
#
OBS_VM_USE_TMPFS=""

## Path:        Applications/OBS
## Description: Memory allocated for each VM (512) if not set
## Type:        integer
## Default:     ""
## Config:      OBS
#
#
OBS_INSTANCE_MEMORY=""

## Path:        Applications/OBS
## Description: Enable storage auto configuration
## Type:        ("yes" | "")
## Default:     ""
## Config:      OBS
#
# WARNING: this may destroy data on your hard disk !
# This is usually only used on mass deployed worker instances
#
OBS_STORAGE_AUTOSETUP="yes"

## Path:        Applications/OBS
## Description: Setup LVM via obsstoragesetup
## Type:        ("take_all" | "use_obs_vg" | "none")
## Default:     "use_obs_vg"
## Config:      OBS
#
# take_all: WARNING: all LVM partitions will be used and all data erased !
# use_obs_vg:  A lvm volume group named "OBS" will be re-setup for the workers.
#
OBS_SETUP_WORKER_PARTITIONS="use_obs_vg"

## Path:        Applications/OBS
## Description: Size in MB when creating LVM partition for cache partition
## Type:        integer
## Default:     ""
## Config:      OBS
#
```

```
#
OBS_WORKER_CACHE_SIZE=""

## Path:        Applications/OBS
## Description: Size in MB when creating LVM partition for each worker root partition
## Type:        integer
## Default:     ""
## Config:      OBS
#
#
OBS_WORKER_ROOT_SIZE=""

## Path:        Applications/OBS
## Description: Size in MB when creating LVM partition for each worker swap partition
## Type:        integer
## Default:     ""
## Config:      OBS
#
#
OBS_WORKER_SWAP_SIZE=""

## Path:        Applications/OBS
## Description: URL to a proxy service for caching binaries used by worker
## Type:        string
## Default:     ""
## Config:      OBS
#
#
OBS_WORKER_BINARIES_PROXY=""

## Path:        Applications/OBS
## Description: URL to a ssh pub key to allow root user login
## Type:        string
## Default:     ""
## Config:      OBS
#
# This is usually used on mass (PXE) deployed workers)
#
OBS_ROOT_SSHD_KEY_URL=""

## Path:        Applications/OBS
## Description: URL to a script to be downloaded and executed
## Type:        string
## Default:     ""
## Config:      OBS
#
# This is a hook for doing special things in your setup at boot time
```

```
#
OBS_WORKER_SCRIPT_URL=""
```

3.1.2.2 BSConfig.pm

This file is a perl module used by most back-end scripts, it mainly defines global variables. Since it is a perl module, after changes the back-end servers need to be restarted to become aware of the changes.



Warning

If there is a Perl syntax error in this file, the services will not start. Most likely you forgot the semicolon on the end of a statement.

TABLE 3.5: BSConfig.pm VARIABLES

Variable	Description	Values `default`	Remarks
$hostname	FQDN of the back-end host		leave as it is
$ip	IP address of the back-end host		leave as it is
$frontend	FQDN of the front-end host	`undef`	set only if the front-end runs on another host
$ipaccess	Map of IP access rules		Add all hosts if partition are used
$srcserver	URL of the source server	`'http://$hostname: 5352'`	
$reposerver	URL of the repo server	`'http://$hostname: 5252'`	partition specific
$serviceserver	URL of the service server	`'http://$hostname: 5152'`	
$workersrcserver	URL of the source server		optional for worker access

Variable	Description	Values `default`	Remarks
$workerreposerver	URL of the repo server		optional for worker access
$clouduploadserver	URL of the cloud upload server	`'http://$hostname: 5452'`	
$servicedir	Path to the service scripts	`/usr/lib/obs/service/`	
$servicetempdir	Path to service temp dir	`/var/tmp/`	optional
$serviceroot	Prefix to servicedir		optional
$service_maxchild	Maximum number of concurrent jobs for source service	integer	unlimited if not set
$gpg_standard_key	Path to the standard sign key		
$hermesserver	URL of the notification server		optional
$hermesnamespace	Namespace for the notifications		optional
$notification _plugin	notification plugins		optional
@reposervers	List of reposervers	`("http://$hostname: 5252")`	
$bsdir	Path to the back-end directory	`/srv/obs`	
$bsuser	OS user running the back-end	`obsrun`	

Variable	Description	Values `default`	Remarks
$bsgroup	OS group running the back-end	`obsrun`	
$bsquotafile	Package quota for projects		optional
$sched_asyncmode	Use asynchronous scheduler		Avoid issues with remote projects on slow networks
$sched_startupmode	Cold start mode	`0` 1 2	
$disable_data_sync	fdatasync		may cause data corruption
$rundir	back-end communication	`$bsdir/run`	
$logdir	log directory	`$bsdir/log`	
$nosharedtrees	Shared trees 0=shared 1=not shared 2=not shared with fallback	0 1 `2`	optional for non-ACL systems, should be set for access control
$packtrack	enable binary release tracking	`[]`	
$limit_projects	limit visibility of projects for some architectures		optional
$relsync_pool	allow separation of releasenumber syncing per architecture		
$stageserver	stage server		rsync URI
$stageserver_sync	Extra stage sync server		rsync URI

Variable	Description	Values `default`	Remarks
$sign	Path to sign script		
$sign_project	call sign with --project <project>	0 `1`	
$keyfile	Global sign key		
$localarch	Local architecture for product building		
$buildlog_maxsize	worker max buildlog size	`'500 * 1000000'`	in bytes
$buildlog_maxidle	Time with no changes in the buildlog will kill the job	`'8 * 3600'`	in sec
$xenstore_maxsize	xenstore size	`'20 * 1000000'`	current XEN has no xenstore anymore
$gettimeout	Max timeout for get	`'1 * 3600'`	in sec
$workerhostcheck	check script for worker		
$powerhosts	Worker with more resources		obsolete use constraints
$powerpkgs	packages which need workers with more resources		obsolete use constraints
$norootexceptions	List of packages need to build as root		
$old_style_services	Use old style source service handling	`0` 1	

Variable	Description	Values `default`	Remarks
$partition	Current partition		see *Section 2.4, "Distributed Setup"*
$partitioning	Partition project mapping		see *Section 2.4, "Distributed Setup"*
$partitionservers	Partition server mapping		see *Section 2.4, "Distributed Setup"*
$dispatch_adjust	Adjust dispatch priority		see *Section 5.5.2, "`dispatch_adjust` Array"*
$publishedhook_use_regex	Use regular expressions in publish hook map	`0` 1	see *Section 5.6, "Publisher Hooks"*
$publishedhook	Publish hook map		see *Section 5.6, "Publisher Hooks"*
$unpublishedhook_use_regex	Use regular expressions in unpublish hook map	`0` 1	see *Section 5.7, "Unpublisher Hooks"*
$unpublishedhook	Unpublish hook map		see *Section 5.7, "Unpublisher Hooks"*

Example BSConfig.pm

```
#
# Copyright (c) 2006, 2007 Michael Schroeder, Novell Inc.
#
# This program is free software; you can redistribute it and/or modify
# it under the terms of the GNU General Public License version 2 as
# published by the Free Software Foundation.
#
# This program is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
# GNU General Public License for more details.
#
# You should have received a copy of the GNU General Public License
# along with this program (see the file COPYING); if not, write to the
# Free Software Foundation, Inc.,
```

```
# 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA
#
################################################################
#
# Open Build Service Configuration
#

package BSConfig;

use Net::Domain;
use Socket;

my $hostname = Net::Domain::hostfqdn() || 'localhost';
# IP corresponding to hostname (only used for $ipaccess); fallback to localhost since
 inet_aton may fail to resolve at shutdown.
my $ip = quotemeta inet_ntoa(inet_aton($hostname) || inet_aton("localhost"));

my $frontend = undef; # FQDN of the Web UI/API server if it's not $hostname

# If defined, restrict access to the backend servers (bs_repserver, bs_srcserver,
 bs_service)
our $ipaccess = {
   '127\..*' => 'rw', # only the localhost can write to the backend
   "^$ip" => 'rw',    # Permit IP of FQDN
   '.*' => 'worker',  # build results can be delivered from any client in the network
};

# IP of the Web UI/API Server (only used for $ipaccess)
if ($frontend) {
  my $frontendip = quotemeta inet_ntoa(inet_aton($frontend) || inet_aton("localhost"));
  $ipaccess->{$frontendip} = 'rw' ; # in dotted.quad format
}

# Also change the SLP reg files in /etc/slp.reg.d/ when you touch hostname or port
our $srcserver = "http://$hostname:5352";
our $reposerver = "http://$hostname:5252";
our $serviceserver = "http://$hostname:5152";

# you can use different ports for worker connections
#our $workersrcserver = "http://$hostname:5353";
#our $workerreposerver = "http://$hostname:5253";

our $servicedir = "/usr/lib/obs/service/";
#our $servicetempdir = "/var/temp/";
#our $serviceroot = "/opt/obs/MyServiceSystem";

# Maximum number of concurrent jobs for source service
```

```
#our $service_maxchild = 20;

our $gpg_standard_key = "/srv/obs/obs-default-gpg.asc";
# optional notification service:
#our $hermesserver = "http://$hostname/hermes";
#our $hermesnamespace = "OBS";
#
# Notification Plugin, multiple plugins supported, separated by space
#our $notification_plugin = "notify_hermes notify_rabbitmq";
#

# For the workers only, it is possible to define multiple repository servers here.
# But only one source server is possible yet.
our @reposervers = ("http://$hostname:5252");

# Package defaults
our $bsdir = '/srv/obs';
our $bsuser = 'obsrun';
our $bsgroup = 'obsrun';
#our $bsquotafile = '/srv/obs/quota.xml';

# Use asynchronus scheduler. This avoids hanging schedulers on remote projects,
# when the network is slow or broken. This will become the default in future
# our $sched_asyncmode = 1;

# Define how the scheduler does a cold start. The default (0) is to request the
# data for all packages, (1) means that only the non-remote packages are fetched,
# (2) means that all of the package data fetches get delayed.
# our $sched_startupmode = 0;

# Disable fdatasync calls, increases the speed, but may lead to data
# corruption on system crash when the filesystem does not guarantees
# data write before rename.
# It is esp. required on XFS filesystem.
# It is safe to be disabled on ext4 and btrfs filesystems.
#our $disable_data_sync = 1;

# Package rc script / backend communication + log files
our $rundir = "$bsdir/run";
our $logdir = "$bsdir/log";

# optional for non-acl systems, should be set for access control
# 0: trees are shared between projects (built-in default)
# 1: trees are not shared (only usable for new installations)
# 2: new trees are not shared, in case of a missing tree the shared
#    location is also tried (package default)
our $nosharedtrees = 2;
```

```
# enable binary release tracking by default for release projects
our $packtrack = [];

# optional: limit visibility of projects for some architectures
#our $limit_projects = {
# "ppc" => [ "openSUSE:Factory", "FATE" ],
# "ppc64" => [ "openSUSE:Factory", "FATE" ],
#};

# optional: allow seperation of releasnumber syncing per architecture
# one counter pool for all ppc architectures, one for i586/x86_64,
# arm archs are separated and one for the rest in this example
our $relsync_pool = {
 "local" => "local",
 "i586" => "i586",
 "x86_64" => "i586",
 "ppc" => "ppc",
 "ppc64" => "ppc",
 "ppc64le" => "ppc",
 "mips" => "mips",
 "mips64" => "mips",
 "mipsel" => "mipsel",
 "mips64el" => "mipsel",
 "aarch64"  => "arm",
 "aarch64_ilp32"  => "arm",
 "armv4l"  => "arm",
 "armv5l"  => "arm",
 "armv6l"  => "arm",
 "armv6hl" => "arm",
 "armv7l"  => "arm",
 "armv7hl" => "arm",
 "armv5el" => "armv5el", # they do not exist
 "armv6el" => "armv6el",
 "armv7el" => "armv7el",
 "armv8el" => "armv8el",
 "sparcv9" => "sparcv9",
 "sparc64" => "sparcv9",
};

#No extra stage server sync
#our $stageserver = 'rsync://127.0.0.1/put-repos-main';
#our $stageserver_sync = 'rsync://127.0.0.1/trigger-repos-sync';

#No package signing server
our $sign = "/usr/bin/sign";
#Extend sign call with project name as argument "--project $NAME"
```

```
#our $sign_project = 1;
#Global sign key
our $keyfile = "/srv/obs/obs-default-gpg.asc";

# Use a special local arch for product building
# our $localarch = "x86_64";

# config options for the bs_worker
#
#our buildlog_maxsize = 500 * 1000000;
#our buildlog_maxidle =   8 * 3600;
#our xenstore_maxsize =  20 * 1000000;
#our gettimeout =         1 * 3600;
#
# run a script to check if the worker is good enough for the job
#our workerhostcheck = 'my_check_script';
#
# Allow to build as root, exceptions per package
# the keys are actually anchored regexes
# our $norootexceptions = { "my_project/my_package" => 1, "openSUSE:Factory.*/
installation-images" => 1 };

# Use old style source service handling
# our $old_style_services = 1;

###
# Optional support to split the binary backend. This can be used on large servers
# to separate projects for better scalability.
# There is still just one source server, but there can be multiple servers which
# run each repserver, schedulers, dispatcher, warden and publisher
#
# This repo service is the 'home' server for all home:* projects. This and the
# $reposerver setting must be different on the binary backend servers.
# our $partition = 'home';
#
# this defines how the projects are split. All home: projects are hosted
# on an own server in this example. Order is important.
# our $partitioning = [ 'home:' => 'home',
#                       '.*'    => 'main',
#                     ];
#
# our $partitionservers = { 'home' => 'http://home-backend-server:5252',
#                           'main' => 'http://main-backend-server:5252',
#                         };

# Publish hooks
our $publishedhook_use_regex = 1;
```

```
our $publishedhook = {
    "Product\/SLES12"     => "/usr/local/bin/script2run_sles12",
    "Product\/SLES11.*"   => "/usr/local/bin/script2run_sles11",
};


# host specific configs
my $hostconfig = __FILE__;
$hostconfig =~ s/[^\/]*$/bsconfig.$hostname/;
if (-r $hostconfig) {
  print STDERR "reading $hostconfig...\n";
  require $hostconfig;
}

1;
```

3.2 Log Files

3.2.1 Front-end

The front-end log files are found under */srv/www/obs/api/log*.

The following front-end log files exist:

- apache_access.log - apache requests
- apache_error.log - errors from apache
- backend_access.log - API → backend requests
- clockworkd.clock.output → timer event log
- delayed_job.log → delayed job log
- production.log→ main ruby log
- production.searchd.log - search daemon log
- production.searchd.query.log - search request logs

3.2.2 Back-end

The back-end log files are found by default under */srv/obs/log/*.

The following back-end log files exist:

- dispatcher.log - dispatcher log
- dodup.log - download on demand log (since 2.7)
- publisher.log - publisher log
- rep_server.log - repo server log
- scheduler_<arch>.log - scheduler log for each architecture
- signer.log - sign service log
- src_server.log - source server log
- src_service.log - source service daemon log
- warden.log - warden log
- clouduploadserver.log - cloud upload server log
- clouduploadworker.log - cloud upload worker log

The following log files for the upload jobs exist inside the */srv/obs/cloudupload* directory (also linked in */bs/cloudupload*):

- <job_id>.log - log files for undone upload jobs
- done/<job_id>.log - log files for already finished upload jobs

3.3 `/srv/obs` Tree

The default back-end data directory is located under */srv/obs/*. Here are a bunch of subdirectories used for communication between the different server, to store data, status information and logs. Here is one file ***configuration.xml*** in the top directory, which stores the global OBS configuration for the back-end. You should not modify this file directly, but use the API /configuration interface instead, since this information needs to kept in sync with the front-end.

3.3.1 `build` Directory

In this subdirectory managed by the repo server daemon, all repository data, metadata and build results are stored in a hierarchical tree.

Example build directory tree of a binary imported distribution (OpenSUSE:13.2) and a small test project with 3 packages:

```
├── openSUSE:13.2
│   └── standard
│       ├── i586
│       │   └── :full
│       └── x86_64
│           └── :full
├── Test1
│   └── os13.2
│       ├── i586
│       │   ├── :full
│       │   ├── :logfiles.fail
│       │   ├── :logfiles.success
│       │   ├── :meta
│       │   ├── :repo
│       │   ├── rsync
│       │   ├── srtp
│       │   └── wget
│       └── x86_64
│           ├── :full
│           ├── :logfiles.fail
│           ├── :logfiles.success
│           ├── :meta
│           ├── :repo
│           ├── rsync
│           ├── srtp
│           └── wget
```

3.3.2 `clouduploa`d Directory

Info for cloud upload jobs is stored here, it has a subdir named *done* for storing the already finished jobs.

3.3.3 db Directory

Back-end database root directory use by the source server, repo server scheduler and publisher. Nobody should touch this.

3.3.4 `diffcache` Directory

Cache for source server compare operations.

3.3.5 `events` Directory

Communication between services.

3.3.6 `info` Directory

Scheduler information managed by the scheduler and used by the repo server.

3.3.7 `jobs` Directory

The build jobs are stored in the /srv/obs/jobs directory. They are organized bybuild architecture:

```
jobs
├── armv7l
├── i586
├── load
└── x86_64
    └── Release:Stable::SLE-12_GA::CI-demo-36db80552b735e193dced13f058f866f
```

The jobs/load file contains statistical data about the build jobs.

3.3.8 `log` Directory

Contains the log files of the back-end daemons.

3.3.9 `projects` Directory

Contains the project hierarchy and metadata under revision control.

3.3.10 `remotecache` Directory

Cache for remote repository information.

3.3.11 `repos` Directory

Directory managed by the publisher to collect build results, also used by the repo server and scheduler to find build results.

3.3.12 `repos_sync` Directory

Directory with files pointing to the project root directories, helper for publisher rsync.

3.3.13 `run` Directory

State and lock information for the back-end daemons

3.3.14 `sources` Directory

All package sources under revision control in one directory per package, managed by the source server. Package sources are by default deduplicated across all projects, as long a source file has the same MD5 sum, it is only stored once. A pseudo ***'_project'*** package exist in the directory containing the project meta-data revisions. ***':service'*** and ***':upload'*** are temporary directories used by the source server.

Example sources directory structure:

```
sources/
├── CI-demo
[...]
├── srtp
├── test1
├── _project
├── :service
└── :upload
```

3.3.15 `trees` Directory

Revision control data for project and packages, managed by the source server.

3.3.16 `upload` Directory

Temporary directory for uploading files for other back-end components.

3.3.17 `workers` Directory

Worker information

3.4 Metadata

3.4.1 OBS Revision Control

This section gives a short generic overview how the revision information are stored in the OBS back-end for packages and projects. The OBS back-end stores all files in a light weight content based hierarchical tree. Each file is hashed (with MD5) and stored with the hash as part of the filename under the */srv/obs/tree* or */srv/obs/sources* directories. The revision information is stored in separate files by the Source Server in the */srv/obs/projects directory*.

3.4.1.1 OBS revision control files

The revision information is stored in simple CSV like file format with a bar (|) as delimiter between the 8 columns. The files do have the extension *.rev* for package/project revision data and *.mref* for meta revision data. The hash then points to a <hash>-MD5SUMS file in the */srv/obs/tree/* directories which have the file list with MD5 hashes of this revision. The hashes in this file list are pointing to the source files in the */srv/obs/sources* tree.

An example revision file:

```
1|1|56cdd3adb778089d1fcc49b92bb93e5b|0.9|1464005086|user4|initial version|
2|2|fe7aa1ade5c9d005de738c234c90bc90|0.9|1464005304|user4|fix spec file|
3|1|72c7986e694f45ab1a62779e64e92a8f|1.0|1464005339|user4|new version|
4|2|699e9931e6f167d78e65bbe5853f592f|1.0|1464006221|user4|add patch file|
5|1|0cfc3a2297f38d2aa9d8d0e98fc22a38|1.1|1464007797|user4|new version|
```

TABLE 3.6: THE 8 COLUMNS

Column	Content	XML tag	may empty
1	revision number	ref	no
2	version revision number	vref	yes

Column	Content	XML tag	may empty
3	hash	srcmd5	no
4	version	version	yes
5	time stamp	time	no
6	user	user	no
7	commit message	comment	yes
8	request id	requestid	yes

Depending on the target (package, project or metadata) used, fields can be empty or have special values, for example, `unknown` for the version.

Example MD5SUMS file

```
/srv/obs # cat trees/Test1/package1/56cdd3adb778089d1fcc49b92bb93e5b-MD5SUMS
0a17daaa913df9e50ee65e83a1898363  package1.spec
1f810b3521242a98333b7bbf6b2b7ef7  test1.sh
```

3.4.1.2 OBS Revision API

The revision info can be retrieved via API calls for the specific package, for example, using `/source/<project>/<package>/_history`.

Specific revisions of files can be retrieved with the optional "rev=N" parameter, for example, */source/<project>/<package>/<file>?rev=N*.

On PUT and POST methods for files the optional "comment=some+comment" can be used to set a commit message.

3.4.2 Project Metadata

Project metadata are XML files containing the meta project information, such as title, description, related user and groups with roles, build settings, repository settings, publish settings, debug settings and more.

TABLE 3.7: PROJECT META XML

XML tag	Attributes	Content
project	name	project name
title		Short description
description		Developer information
person	userid	login name
person	role	role (maintainer, bugowner, …)
group	groupid	group name
group	role	role (maintainer, bugowner, …)
devel		An optional devel project
build		optional build flags
publish		optional publish flags
useforbuild		optional useforbuild flags
debuginfo		optional debuginfo flags
binarydownload		optional binarydownload flags
repository	name	name of the repository for build results
repository path	project	name of the source project for remaining build requires
repository path	repository	name of repository in the source project
repository arch		architecture name
remoteurl		path to a remote OBS API for interconnect

Example project metadata:

```
<project name="Test11">
```

```
  <title>Test project 11</title>
  <description>Project for demo</description>
  <person userid="Admin" role="maintainer"/>
  <person userid="user0" role="maintainer"/>
  <group groupid="obsprj3" role="maintainer"/>
  <repository name="openSUSE_13.2">
    <path project="openSUSE.org:openSUSE:13.2" repository="standard"/>
    <arch>x86_64</arch>
  </repository>
</project>
```

3.4.3 Package Metadata

XML file about package meta information, like Title, description, related user and groups with roles, build settings, publish settings, debug settings and more. Most XML tags are the same as for projects.

Example package metadata:

```
<package name="test1" project="Test11">
  <title>A test package for learning</title>
  <description>An example test package for learning.&#13;
</description>
  <person userid="user5" role="bugowner"/>
  <person userid="user5" role="maintainer"/>
  <build>
    <enable repository="openSUSE_13.2"/>
  </build>
  <publish>
    <disable repository="openSUSE_13.2"/>
  </publish>
  <debuginfo>
    <disable/>
  </debuginfo>
</package>
```

3.4.4 Attribute Metadata

Attributes can be used to add special information to packages. Attributes can be used to trigger special actions.

Example attribute data:

```
<attributes>
  <attribute name="Issues" namespace="OBS"/>
```

```
  <attribute name="AutoCleanup" namespace="OBS">
    <value>2016-06-30 00:00:00</value>
  </attribute>
  <attribute name="AutoCleanup" namespace="OBS">
    <value></value>
  </attribute>
</attributes>
```

3.4.5 Job Files

Jobs are stored by the scheduler in the `/srv/obs/jobs` directory and contain the build setup information for the package, for example, a reference to the exact source version, build dependencies, build repository information, timestamps.

Sample job file:

```
<buildinfo project="Release:Stable" repository="SLE-12_GA" package="CI-demo"
srcserver="http://obs.b1-systems.de:5352"
reposerver="http://obs.b1-systems.de:5252">
  <job>Release:Stable::SLE-12_GA::
 CI-demo-36db80552b735e193dced13f058f866f</job>
  <arch>x86_64</arch>
  <srcmd5>36db80552b735e193dced13f058f866f</srcmd5>
  <verifymd5>36db80552b735e193dced13f058f866f</verifymd5>
  <rev>2</rev>
  <disturl>obs://b1-systems.de/Release:Stable/SLE-12_GA/
  36db80552b735e193dced13f058f866f-CI-demo</disturl>
  <reason>new build</reason>
  <needed>0</needed>
  <revtime>1461077600</revtime>
  <readytime>1461077708</readytime>
  <file>CI-demo.spec</file>
  <versrel>0.1.9-2</versrel>
  <bcnt>1</bcnt>
  <release>2.1</release>
  <debuginfo>1</debuginfo>
  <prjconfconstraint>linux:version:min 3.0.0</prjconfconstraint>
  <bdep name="aaa_base" preinstall="1" runscripts="1" notmeta="1" />
  <bdep name="attr" preinstall="1" notmeta="1" />
  <bdep name="bash" preinstall="1" notmeta="1" />
  <bdep name="coreutils" preinstall="1" notmeta="1" />
  <bdep name="diffutils" preinstall="1" notmeta="1" />
  <bdep name="filesystem" preinstall="1" notmeta="1" />
  <bdep name="fillup" preinstall="1" notmeta="1" />
  <bdep name="glibc" preinstall="1" notmeta="1" />
```

```
  <bdep name="grep" preinstall="1" notmeta="1" />
  <bdep name="libbz2-1" preinstall="1" notmeta="1" />
  <bdep name="libgcc_s1" preinstall="1" notmeta="1" />
  <bdep name="m4" preinstall="1" notmeta="1" />
[...]
  <path project="Release:Stable" repository="SLE-12_GA"
 server="http://obs.b1-systems.de:5252" />
  <path project="SUSE:SLE-12:GA" repository="standard"
 server="http://obs.b1-systems.de:5252" />
</buildinfo>
```

4 Security Concepts

4.1 General Paradigm

The general paradigm of Open Build Service is to host all content on its own. Every part required to rebuild a package is hosted in Open Build Service to guarantee reproducibility. This includes the ability to rebuild the build environment. However, optional services to integrate remote resources exist as well. These resources are either mirrored and stored in revision control system or just cached.

4.1.1 Frontend

The API and web interface frontends is the only part which must be accessible from public network. A SSL/TLS certificate is highly recommended.

4.1.1.1 Access to Mirror Servers

The following services require access to stage servers. These servers can be used to publish content without the need to make Open Build Service server parts public available.

- The publisher is used to publish any build results to the stage servers. It needs access for rsync service to the servers.
- The source publisher can be used similar to publish sources belonging to published build results (optional service).

4.1.1.2 Access to the Public Network

The following services may require access to the public network.

- The srcserver when using the interconnect functionality to use content from remote Open Build Service instances.
- obsdodup when using external repositories (optional service). The obssrcserver would need access then as well for downloading content.

- obssourceservice when supporting automatic source procession (optional service) from remote. This service may be used to download content (for example, Git repositories) as part of the sources.
- obsclouduploadworker when publishing to public cloud instances is wanted.

4.1.1.3 Worker network

It is recommended to run the Open Build Service workers in an isolated network. This is an additional security mechanism in case of a security breach on a worker. This network needs access to the source and repository servers of the Open Build Service backend, but nowhere else.

4.1.1.4 Signer network

It is recommended to run the signd on an isolated host. The signer services need to stay on the Open Build Service backend servers, they are just used for scheduling sign jobs. The signd is the critical instance which hosts the single instance signature key. All further keys which belong to any project are created by the signd, crypted with the instance key and delivered to the backend servers. That way the signd instance is stateless after initial setup and it is enough to backup the backend servers. Any sign job sends the encrypted private key to the signd server which is decrypting it for signing the content. Access to the signd server must be limited to Open Build Service backend server components (source server and publisher).

4.1.2 Build Environment

The build environment is created by obsworker instances via the build script. Inside the build instances unverified and potentially harmful code is executed. Given that a user can run also any provided kernel the isolation must happen on VM layer. The only supported VM types, which are considered as secure are KVM and XEN VMs. The VMs prohibit any network access from the running instances. The build script is always creating a new file system, copies in all required data and executes the VM.

The build results get extracted directly from the block device. This is done to avoid mounting it which could be used to breach the host kernel in the file system code by crafting the used filesystem during build.

Every build also stores the used binaries inside of the `_buildenv` file. This can be used to re-create the build environment later, even when newer updates got released meanwhile. This requires however a maintenance setup to avoid that former releases get removed.

The build tools (like rpmbuild) are running usually as non-root user, but this is not a security instrument. It is only a quality mechanic to ensure that source packages are rebuildable without root permissions.

4.1.3 Source Revision System

The source revision storage system is part of Open Build Service. The identification of sources still happens using MD5 sums for historic reasons. MD5 is considered to be still good enough for identifying a source, but it is known to be attackable. Recent versions of the osc command line tooling is sending therefore also a SHA256 sum in addition which is used to detect collisions by the source server.

Sources can reference other package sources via `_link` files. These can be pinned to a revision or to always using the latest one. That way underlying changes get merged automatically. This is useful for shared development and automated builds, but it should be avoided for base projects.

4.1.4 Permission Handling

Authorization for write operations is done via the maintainer role on package or project level. On project level the projects are organized in namespaces which are defined via colons inside the project name. A maintainer role on a higher namespace grants the permission to claim maintainer role on any deeper project. Official projects should therefore be organized under their own top level namespaces (for example, openSUSE: namespace in our reference instance).

4.1.5 Signature Handling

Signatures are used to proof the origin of a shipment independent of Open Build Service instance. Once the `signd` daemon has been enabled by the administrator, any binary result like RPM packages, images, containers, or meta data gets signed. All Open Build Service instances have a master key to sign results. This master key has two purposes: First, it is used for signing if a dedicated signing key can not be found. Second, it is used to encrypt dedicated project signing keys so that they can be stored inside of the projects instead of on the signing server. If a project does not have a signing key, the build server will search the parent projects. This follows the logic of the write permission handling.

An Open Build Service user (including the Open Build Service admin user) has only limited options to deal with the key setup. The user can create, delete, or extend the expire time of a key. It is not possible to import an external key. This guarantees that no key can be used outside of the Open Build Service instance to sign Open Build Service content.

A SSL certificate is derived from the keys when needed (for example, for secure boot).

GPG keypairs are created by the `signd` daemon. Therefore, it is recommend to run this daemon on a separated and protected host. The master keypair should exist only on this `signd` host. Any further created keypair is not stored on the `signd` instance, instead the private part is encrypted with the master key

and transferred to the Open Build Service backend. Thus, the `signd` instance is stateless and needs no recurring backups after initial setup. All keypairs (public and private parts) are therefore part of the backup of the Open Build Service backend servers. The `sign` executable transfers just the hash to be signed (not the entire file content) together with the encrypted private key to `signd`. The daemon decrypts the private key with the master key, creates the signature, and sends it back to the client. The returned signature is applied to the binary by the `sign` executable afterwards.

A compromised backend would still result in a serious security incident since any content can be signed with any project key. The private keys are not compromised themselves though.

4.2 Trust Zones

Open Build Service (OBS) components deal with different trust zones. These are separated via network or virtualization mechanics.

4.2.1 Public Zones

Public zones are areas where any code under user control is running.

4.2.1.1 External Network

This can be the public Internet if the Open Build Service instance is a public instance. Requests can only be triggered via http, secured by SSL to the API instance here. Every developer is using this interface to submit changes. However, further components may open connections to the Internet as described below.

4.2.1.2 Untrusted Code

All code which is used to build content is considered to be untrusted code. This includes even the Linux kernel, since users can build and run their own kernels. The security layer here is usually either KVM and XEN virtualization. The instances are running without any network enabled. Build results get copied by reading a simple blocklist directly from the device. A security issue in the file system code can therefore not be used to compromise the worker.

4.2.2 Demilitarized Zone (DMZ)

The Demilitarized Zone contains services which interact with the public zone directly.

4.2.2.1 Open Build Service Frontend

The frontend service is the only service which provides an open port. It implements the authorization of user requests. The authentication may happen in an external service like a proxy or ldap server. The frontend reads and writes changes to the database and the source server only.

4.2.2.2 Open Build Service Frontend Background Services

Open Build Service frontend background services handle less time critical operations. This includes services which read data from external services like bug trackers, sending notifications or long running jobs.

4.2.2.3 Stage Server

The stage server is providing the public content of the Open Build Service backends. The server can be publicly accessible or just an upload server to a mirror infrastructure.

4.2.2.4 Cloud Uploader

The cloud uploader is uploading build results on user request. It reads from the binary servers and sends the content via cloud specific plugins to external instances. This is an optional service.

4.2.2.5 Source Service Server

The source service server is acting based on uploaded sources. The services should be written with security in mind and the administrator can decide which services are trustworthy. Official releases of Open Build Service define a minimum set of especially trusted services which received a security review. A container can be used for additional security for each service run.

4.2.3 Internal Zone

The internal zone is running service which are supposed to work without further external dependency.

4.2.3.1 Open Build Service Source Server

The source server coordinates changes to package and project configuration. In addition to that all events between the binary backends and notifications to the frontend get synchronized. Any source change is tracked and stored in the revision history. There can only be a single source server per OBS install.

4.2.3.2 Open Build Service Binary Servers

Binary Servers are hosting all content of build results. They also prepare public repositories and deliver them to the staging server.

4.2.3.3 External Dependencies

The internal zone has no external dependency. However, the administrator may decide to run the Open Build Service instance depending on external OBS instances or on external repositories. In that case the internal zone is downloading content from the resources specified by the administrator.

4.2.4 Worker Zone

The Open Build Service workers are running in an own isolated network. They access only source and binary servers from internal zone.

4.2.5 Signing Server

The signing server is supposed to be the most isolated service. It is supposed to be stateless after initial setup. Avoid to enable any remote access.

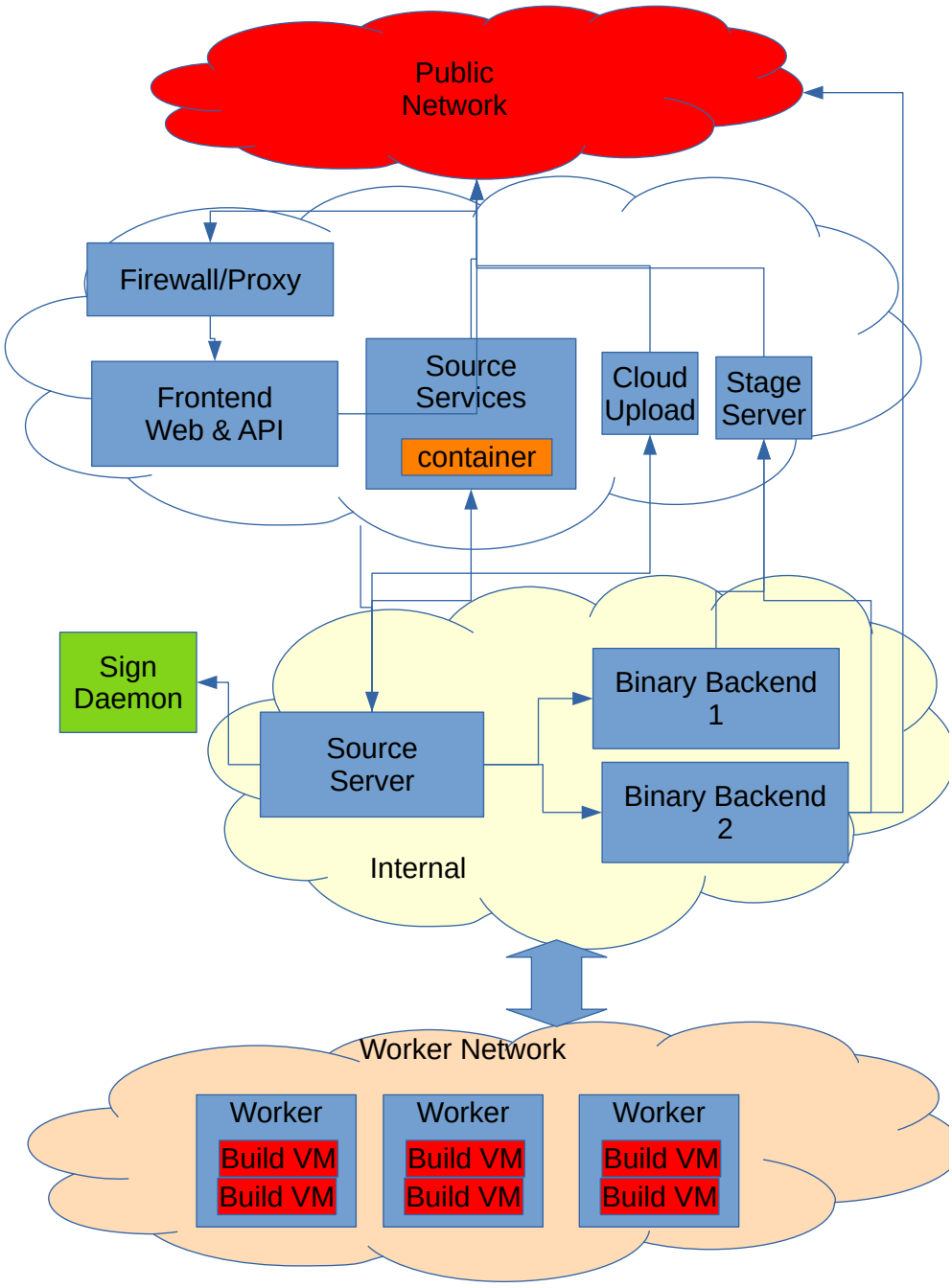


FIGURE 4.1: TRUST ZONES OF OPEN BUILD SERVICE

5 Administration

5.1 Tools

5.1.1 `obs_admin`

obs_admin is a command-line tool used on the back-end server(s) to manage running services, submit maintenance tasks, and debug problems. It should be only used by experienced admins.

It has built-in help which you can display with *obs_admin --help*.

Options to control the running services:

```
Job Controlling
===============

 --shutdown-scheduler <architecture>
   Stops the scheduler nicely with dumping out its current state
   for fast startup.

 --check-project <project> <architecture>
 --check-project <project> <repository> <architecture>
 --check-all-projects <architecture>
   Check status of a project and its repositories again

 --deep-check-project <project> <architecture>
 --deep-check-project <project> <repository> <architecture>
   Check status of a project and its repositories again
   This deep check also includes the sources, in case of lost events.

 --check-package <project> <package> <architecture>
   Check status of a package in all repositories

 --publish-repository <project> <repository>
   Creates an event for the publisher. The scheduler is NOT scanning for new packages.
   The publisher may skip the event, if nothing has changed.
   Use --republish-repository when you want to enforce a publish.

 --unpublish-repository <project> <repository>
   Removes the prepared :repo collection and let the publisher remove the result. This
   is also updating the search database.
   WARNING: this works also for locked projects!
```

```
 --prefer-publish-event <name>
   prefers a publish event to be next. <name> is the file name inside of the publish
   event directory.

 --republish-repository <project> <repository>
   enforce to publish a repository

 --rebuild-full-tree <project> <repository> <arch>
   rebuild the content of :full/ directory

 --clone-repository <source project> <source repository> <destination repository>
 --clone-repository <source project> <source repository> <destination project>
 <destination repository>
   Clone an existing repo into another existing repository.
   Usefull for creating snapshots.

 --rescan-repository <project> <repository> <architecture>
   Asks the scheduler to scan a repository for new packages and add
   them to the cache file.

 --force-check-project <project> <repository> <architecture>
   Enforces the check of an repository, even when it is currently blocked due to amount
 of
   calculating time.

 --create-patchinfo-from-updateinfo
   creates a patchinfo submission based on an updateinfo information.
```

Options for maintenance are:

```
Maintenance Tasks
=================

Note: the --update-*-db calls are usually only needed when corrupt data has been created,
 for
      example after a file system corruption.

 --update-source-db [<project>]
   Update the index for all source files.

 --update-request-db
   Updates the index for all requests.

 --remove-old-sources <days> <y> (--debug)
   WARNING: this is an experimental feature atm. It may trash your data, but you have
 anyway
            a backup, right?
   remove sources older than <x> days, but keep <y> number of revisions
```

```
   --debug for debug output
```

Options for debugging:

```
Debug Options
=============

 --dump-cache <project> <repository> <architecture>
   Dumps out the content of a binary cache file.
   This shows all the content of a repository, including all provides
   and requires.

 --dump-state <architecture>

 --dump-project-from-state <project> <arch>
   dump the state of a project.

 --dump-relsync <file>
   To dump content of :relsync files.

 --set-relsync <file> <key> <value>
   Modify key content in a a :relsync file.

 --check-meta-xml <project>
 --check-meta-xml <project> <package>
   Is parsing a project or package xml file and puts out error messages, in case of
 errors.

 --check-product-xml <file>
   Is parsing a product xml file and puts out error messages, in case of errors.
   It does expand all xi:include references and validates the result.

 --check-product-group-xml <file>
   Is parsing a group xml file from a product definition and puts out error messages, in
 case of errors.

 --check-kiwi-xml <file>
 --check-kiwi-xml <project> <package>
   Is parsing a KIWI xml file and puts out error messages, in case of errors.

 --check-constraints <file>
 --check-constraints <project> <package>
   Validates a _constraints file

 --check-pattern-xml <file>
   Is parsing a pattern xml file and puts out error messages, in case of errors.

 --check-request-xml <file>
```

```
  Is parsing a request xml file and puts out error messages, in case of errors.

--parse-build-desc <file> [<arch> [<buildconfigfile>]]
  Parse a spec, dsc or KIWI file with the Build script parser.

--show-scheduler-architectures
  Show all architectures which are configured in configuration.xml to be supported by
this instance.

--show-delta-file <file>
  Show all instructions of a OBS delta file

--show-delta-store <file>
  Show delta store statistics
```

5.1.2 osc

The ***osc*** command-line client is mainly used by developers and packagers. But for some tasks, admin people also need this tool. It too has builtin help: use *osc --help*. The tool needs to be configured first to know the OBS API URL and your user details.

To configure the ***osc*** tool the first time you need to call it with

```
osc -A <URL to the OBS API>
For example:
osc -A https://api.testobs.org
```

Follow the instructions on the terminal.



Warning

The password is stored in clear text in the .oscrc file by default, so you need to give this file restrictive access rights, only read/write access for your user should be allowed. **osc** allows to store the password in other ways (in keyrings for example) and may use different methods for authentication like Kerberos see *Section 5.8.7.2, "Kerberos"*

For admins the most important ***osc*** subcommands are:

- meta - to create or update projects or package data
- API - to read and write online configuration data

5.1.2.1 `osc meta` Subcommand

The **`osc meta`** subcommand is documented inside the "osc" tool itself. This documentation can be displayed by issuing the command:

```
osc meta --help
```

5.1.2.2 `osc api` Subcommand

The **`osc api`** subcommand is documented inside the "osc" tool itself. This documentation can be displayed by issuing the command:

```
osc api --help
```

The OBS API itself is documented at https://api.opensuse.org/apidocs/

Some examples for admin stuff:

```
# Read the global configuration file
  osc api /configuration
# Update the global configuration
  osc api /configuration -T /tmp/configuration.xml

# Read the distributions list
  osc api /distributions
# Udate the distributions list
  osc api /distributions -T /tmp/distributions.xml

# retrieve statistics
  osc api /statistics/latest_added
```

5.2 Managing Build Targets

5.2.1 Interconnect

Using another Open Build Service as source for build targets is the easiest way to start. The advantage is, that you save local resources and you do not need to build everything from scratch. The disadvantage is that you depend on the remote instance, if it has a downtime your instance cannot do any builds for these targets, if the remote admins decide to remove some targets you cannot use them anymore.

The easiest way to interconnect with some of the public OBS instances is to use the Web UI. You need to log in with an administrator account of your instance to do this. On the start page of an administrator account you will find a **Configuration** link. On the Configuration page you find an Interconnect tab on the top, use this and select the public side you want.

If you want to connect to a not listed instance, you can simple create a remote project using the *osc meta prj* command. A remote project differs from a local project as it has a **remoteurl** tag (see *Section 3.4.2, "Project Metadata"*).

Example:

```
<project name="openSUSE.org">
  <title>openSUSE.org Project Link</title>
  <description>
This project refers to projects hosted on the openSUSE Build Service
</description>
  <remoteurl>https://api.opensuse.org/public</remoteurl>
</project>
```

Sending this via osc to the server:

```
osc meta prj -m "add openSUSE.org remote" -F /tmp/openSUSE.org.prj
```

5.2.2 Importing Distributions

FIXME: describe how to do it using DoD

5.3 Source Services

Source Services are tools to validate, generate or modify sources in a trustable way. They are designed as smallest possible tools and can be combined following the powerful idea of the classic UNIX design.

Design goals of source services were:

- server side generated files must be easy to identify and must not be modifiable by the user. This way other users can trust them to be generated in the documented way without modifications.
- generated files must never create merge conflicts
- generated files must be a separate commit to the user change

- services must be runnable at any time without user commit
- services must be runnable on server and client side in the same way
- services must be designed in a safe way. A source checkout and service run must never harm the system of a user.
- services shall be designed in a way to avoid unnecessary commits. This means there shall be no time-dependent changes. In case the package already contains the same file, the newly generated file must be dropped.
- local services can be added and used by everybody.
- server side services must be installed by the admin of the OBS server.
- services can be defined per package or project wide.

5.3.1 Using Services for Validation

Source Services may be used to validate sources. This can happen per package, which is useful when the packager wants to validate that downloaded sources are really from the original maintainer. Or validation can happen for an entire project to apply general policies. These services cannot get skipped in any package

Validation can happen by validating files (for example using the **`verify_file`** or **`source_validator`** service. These services just fail in the error case which leads to the build state "broken". Or validation can happen by redoing a certain action and store the result as new file as **`download_files`** is doing. In this case the newly generated file will be used instead of the committed one during build.

5.3.2 Different Modes When Using Services

Each service can be used in a special mode defining when it should run and how to use the result. This can be done per package or globally for an entire project.

5.3.2.1 Default Mode

The default mode of a service is to always run after each commit on the server side and locally before every local build.

5.3.2.2 `trylocal` Mode

The trylocal mode is running the service locally when using current osc versions. The result gets committed as standard files and not named with _service: prefix. Additionally the service runs on the server by default, but usually the service should detect that the result is the same and skip the generated files. If they differ (for example, because the Web UI or API was used), they are generated and added on the server.

5.3.2.3 `localonly` Mode

The localonly mode is running the service locally when using current osc versions. The result gets committed as standard files and not named with _service: prefix. The service is never running on the server side. It is also not possible to trigger it manually.

5.3.2.4 `serveronly` Mode

The serviceonly mode is running the service on the server only. This can be useful, when the service is not available or can not work on developer workstations.

5.3.2.5 `buildtime` Mode

The service is running inside of the build job, for local and server side builds. A side effect is that the service package is becoming a build dependency and must be available. Every user can provide and use a service this way in their projects. The generated sources are not part of the source repository, but part of the generated source packages. Network access is not be available when the workers are running in a secure mode.

5.3.2.6 `disabled` Mode

The disabled mode is neither running the service locally or on the server side. It can be used to temporarily disable the service but keeping the definition as part of the service definition. Or it can be used to define the way how to generate the sources and doing so by manually calling **`osc service runall`** The result will get committed as standard files again.

5.3.3 Storage of Source Service Definitions

The called services are always defined in a **`_service`** file. It is either part of the package sources or used project-wide when stored inside the **`_project`** package.

The _service file contains a list of services which get called in this order. Each service may define a list of parameters and a mode. The project wide services get called after the per package defined services. The _service file is an xml file like this example:

```
<services>
  <service name="download_files" mode="trylocal" />
  <service name="verify_file">
    <param name="file">krabber-1.0.tar.gz</param>
    <param name="verifier">sha256</param>
    <param
 name="checksum">7f535a96a834b31ba2201a90c4d365990785dead92be02d4cf846713be938b78</param>
  </service>
</services>
```

This example downloads the files via download_files service via the given URLs from the spec file. When using osc this file gets committed as part of the commit. Afterwards the krabber-1.0.tar.gz file will always be compared with the sha256 checksum.

5.3.4 Dropping a Source Service Again

Sometimes it is useful to continue working on generated files manually. In this situation the _service file needs to be dropped, but all generated files need to be committed as standard files. The OBS provides the "mergeservice" command for this. It can also be used via osc by calling **`osc service merge`**.

5.4 Source Publisher

The job of the source publish service is to publish all sources for directly before published binaries. This will include the sources of repackaged binaries. For example, the sources of RPMs which are used inside of product, appliance or container images. A prerequisite for this is that OBS has enabled content tracking for the used projects.

5.4.1 Configuring Source Publisher

The source publishing can be configured via the file */usr/lib/obs/server/BSConfig.pm*, where it can be enabled globally or just for some projects. It is possible to use regular expressions here.

Publishing can be enabled by defining the rsync module to push the content:

```
our $sourcepublish_sync = 'rsync://your_rsync_server/rsync_module';
```

By default every project get published, but it is possible to define a whitelist via:

```
our $sourcepublish_filter = [ "openSUSE:.*", "SUSE:.*" ];
```

5.4.2 Considerations

The source publishing service is publishing the sources as they are hosted in Open Build Service. This means these are the unprocessed sources and the content is not identical to the content of source RPMs for example. Instead these are the sources which are the base for source RPMs.

As a consequence hints like NoSource: tags in spec files are ignored. The only way to disable publishing for the user is to disable access or sourceaccess via the flags.

The filesystem structure is *$project/$package/$srcmd5/*. A inside of binary builds can be used to find the right sources.

Open Build Service will care for de-duplication on the rsync server. This must get implemented there.

5.5 Dispatch Priorities

The dispatcher takes a job from the scheduler and assign it to a free worker. It tries to share the available build time fair between all the project repositories with pending jobs. To achieve this the dispatcher calculates a ***load*** per project repository of the used build time (similar to the system load in Unix operating systems). The dispatcher assigned jobs to build clients from the repository with the lowest load (thereby increasing its load). It is possible to tweak this mechanism via dispatching priorities assigned to the repositories via the ***/build/_dispatchprios*** **API** call or via the ***dispatch_adjust*** array in the *BSConfig.pm* (see *Section 3.1.2.2, "BSConfig.pm"*) configuration file.

5.5.1 The /build/_dispatchprios API Call

The */build/_dispatchprios* API call allows an Admin to set a priority for defined projects and repositories using the HTTP put method. With the HTTP get method the current XML priority file can be read.

```
<dispatchprios>
  <prio project="ProjectName" repository="RepoName" arch="Architecture" adjust="Number" /
>
</dispatchprios>
```

The attributes *project*, *repository* and *arch* are all optional, if for example *arch* and *repository* are missing the entry is used for all repositories and architectures for the given project. It is not supported to use regular expressions for the names. The adjust value is taken as logarithmic scale factor to the current load of the repositories during the compare. Projects without any entry get a default priority of 0, higher values cause the matching projects to get more build time.

Example dispatchprios XML file

```
<dispatchprios>
   <prio project="DemoProject1" repository="openSUSE_Leap_42.1" adjust="10" />
   <prio project="Test1" adjust="5" />
   <prio project="Test11" repository="openSUSE_13.2" arch="i586" adjust="-10"/>
</dispatchprios>
```

TABLE 5.1: ROUNDED SCALE FACTORS RESULTING FROM A PRIORITY

priority	scale factor	priority	scale factor
-50	100000	3	0.5
-30	1000	5	0.3
-20	100	7	0.2
-15	30	10	0.1
-10	10	15	0.03
-7	5	20	0.01
-5	3	30	0.001
-3	2	40	0.0001

priority	scale factor	priority	scale factor
0	1	50	0.00001

5.5.2 `dispatch_adjust` Array

With the ***dispatch_adjust*** array in the *BSConfig.pm* file the dispatch priorities of project repositories based on regular expressions for the project, repository name and maybe architecture. Each match will add or subtract a value to the priority of the repository. The default priority is 0, higher values cause the matching projects to get more build time.

Each entry in the dispatch_adjust array has the format

```
'regex string'  => priority adjustment
```

The full name of a build repository looks like

```
Project:Subproject/Repository/Architecture

Examples:
   Devel:Science/SLES-11/i586
   home:king:test/Leap42/x86_64
```

If a repository matches a string the adjustment is added to the current value. The final value is the sum of the adjustments of all matched entries. This sum is the same logarithmic scale factor as described in the previous section.

Example dispatch_adjust definition in the BSConfig.pm

```
our $dispatch_adjust = [
    'Devel:' => 7,
    'HotFix:' => +20,
    '.+:test.*' => -10,
    'home:' => -3,
    'home:king' => +30,
    '.+/SLE12-SP2' => -40,
];
```

The above example could have the following background: All Devel projects should get some higher priority so the developer jobs getting more build time. The projects under HotFix are very important fixes for customers and so they should get a worker as soon as possible. All projects with test in the name get some penalty, also home projects are getting only about half of the build time as a normal project, with the exception of the home project from king, the user account of the boss. The SLES12-SP2 repository is not in real use yet, but if here is nothing else to do, build for it as well.



Important

The dispatcher calculates the values form the ***'dispatch_adjust'*** array first, if the same project and repository also has an entry in the dispatchprios XML file, the XML file entry will overwrite the calculated priority. The best practice is to only use one of the methods.

5.6 Publisher Hooks

The job of the publisher service is to publish the built packages and/or images by creating repositories that are made available through a web server.

It can be configured to use custom scripts to copy the build results to different servers or do anything with them that comes to mind. These scripts are called ***publisher hooks***.

5.6.1 Configuring Publisher Hooks

Hooks are configured via the configuration file */usr/lib/obs/server/BSConfig.pm*, where one script per project is linked to the repository that should be run if the project/repository combination is published. It is possible to use regular expressions here.

The script is called by the user `obsrun` with the following parameters:

1. information about the project and its repository (for example, `training/SLE11-SP1`)
2. path to published repository (for example, `/srv/obs/repos/training/SLE11-SP1`)
3. changed packages (for example, `x86_64/test.rpm x86_64/utils.rpm`)

The hooks are configured by adding a hash reference named *$publishedhook* to the *BSConfig.pm* configuration file. The key contains the project, and the value references the accompanying script. If the value is written as an array reference it is possible to call the hook with self-defined parameters.

The publisher will add the 3 listed parameters at the end, after the self-defined parameters (in `/usr/lib/obs/server/BSConfig.pm`):

```
our $publishedhook = {
 "Product/SLES12"     => "/usr/local/bin/script2run_sles12",
 "Product/SLES11-SP3" => "/usr/local/bin/script2run_sles11",
 "Product/SLES11-SP4" => "/usr/local/bin/script2run_sles11",
};
```

Regular expressions or substrings can be used to define a script for more than one repository in one project. The use of regular expressions has to be activated by defining `$publishedhook use regex = 1;` as follows (in `/usr/lib/obs/server/BSConfig.pm`):

```
our $publishedhook_use_regex = 1;
our $publishedhook = {
    "Product\/SLES12"     => "/usr/local/bin/script2run_sles12",
    "Product\/SLES11.*"   => "/usr/local/bin/script2run_sles11",
};
```

With self defined parameters:

```
our $publishedhook_use_regex = 1;
our $publishedhook = {
    "Product\/SLES11.*" => ["/usr/local/bin/script2run", "sles11", "/srv/www/
public_mirror"],
};
```

The configuration is read by the publisher at startup only, so it has to be restarted after configuration changes have been made. The hook script's output is not logged by the publisher and should be written to a log file by the script itself. In case of a broken script,this is logged in the publisher's log file (/srv/obs/log/publisher.log by default):

```
Mon Mar  7 14:34:17 2016 publishing Product/SLES12
    fetched 0 patterns
    running createrepo
    calling published hook /usr/local/bin/script2run_sles12
    /usr/local/bin/script2run_sles12 failed: 65280
    syncing database (6 ops)
```

Interactive scripts are not working and will fail immediately.

If you need to do a lot of work in the hook script and do not want to block the publisher all the time, you should consider using a separate daemon that does all the work and just gets triggered by the configured hook script.

The scripts are called without a timeout.

5.6.2 Example Publisher Scripts

5.6.2.1 Simple Publisher Hook

The following example script ignores the packages that have changed and copies all RPMs from the repository directory to a target directory:

```
#!/bin/bash
OBSHOME="/srv/obs"
SRC_REPO_DIR="$OBSHOME/repos"
LOGFILE="$OBSHOME/log/reposync.log"
DST_REPO_DIR="/srv/repo-mirror"
# Global substitution! To handle strings like Foo:Bar:testing - two
#+double-colons!
PRJ_PATH=${1//:/:\/}
PATH_TO_REPO=$2
rsync -a --log-file=$LOGFILE --mkpath $PATH_TO_REPO/ $DST_REPO_DIR/$PRJ_PATH/
```

For testing purposes, it can be invoked as follows:

```
$ sudo -u obsrun /usr/local/bin/publish-hook.sh Product/SLES11-SP1 \
    /srv/obs/repos/Product/SLE11-SP1
```

5.6.2.2 Advanced Publisher Hook

The following example script reads the destination path from a parameter that is configured with the hook script:

```
#!/bin/bash
LOGFILE="/srv/obs/log/reposync.log"
DST_REPO_DIR=$1
# Global substion! To handle strings like Foo:Bar:testing - two
#+double-colons!
PRJ_PATH=${2//:/:\/}
PATH_TO_REPO=$3
mkdir -p $DST_REPO_DIR/$PRJ_PATH
rsync -a --log-file=$LOGFILE --mkpath $PATH_TO_REPO/ $DST_REPO_DIR/$PRJ_PATH/
```

For testing purposes, it can be invoked as follows:

```
$ sudo -u obsrun /usr/local/bin/publish-hook.sh \
    /srv/www/public_mirror/Product/SLES11-SP1 \
```

```
    /srv/obs/repos/Product/SLE11SP1
```

The following example script only copies packages that have changed, but does not delete packages that have been removed:

```
#!/bin/bash

DST_REPO_DIR=$1
PRJ_PATH=${2//:/:\/}
PATH_TO_REPO=$3
shift 3

mkdir -p $DST_REPO_DIR/$PRJ_PATH

while [ $# -gt 0 ]
do
  dir=(${1//\// })
  if [ ! -d  "$DST_REPO_DIR/$PRJ_PATH/$dir" ]; then
    mkdir -p $DST_REPO_DIR/$PRJ_PATH/$dir
  fi
  cp $PATH_TO_REPO/$1 $DST_REPO_DIR/$PRJ_PATH/$1
  shift
done

createrepo $DST_REPO_DIR/$PRJ_PATH/.
```

For testing purposes, it can be invoked as follows:

```
$ sudo -o obsrun /usr/local/bin/publish-hook.sh  /srv/www/public_mirror \
   Product/SLES11-SP1 /srv/obs/repos/Product/SLE11-SP1 \
   src/icinga-1.13.3-1.3.src.rpm x86_64/icinga-1.13.3-1.3.x86_64.rpm \
   x86_64/icinga-devel-1.13.3-1.3.x86_64.rpm
```

5.7 Unpublisher Hooks

The job of the publisher service is to publish the built packages and/or images by creating repositories that are made available through a web server.

The OBS Publisher can be configured to use custom scripts to be called whenever already published packages get removed. These scripts are called **unpublisher hooks**. **Unpublisher hooks** are run before the **publisher hooks**.

5.7.1 Configuring Unpublisher Hooks

Hooks are configured via the configuration file */usr/lib/obs/server/BSConfig.pm*, where one script per project is linked to the repository that should be run if the project/repository combination is removed. It is possible to use regular expressions here.

The script is called by the user `obsrun` with the following parameters:

1. information about the project and its repository (for example, *training/SLE11-SP1*)
2. repository path (for example, `/srv/obs/repos/training/SLE11-SP1`)
3. removed packages (for example, `x86_64/test.rpm x86_64/utils.rpm`)

The hooks are configured by adding a hash reference named *$unpublishedhook* to the *BSConfig.pm* configuration file. The key contains the project and the value references the accompanying script. If the value is written as an array reference, it is possible to call the hook with custom parameters.

The publisher adds the three listed parameters at the end, directly after the custom parameters (in `/usr/lib/obs/server/BSConfig.pm`):

```
our $unpublishedhook = {
    "Product/SLES12"     => "/usr/local/bin/script2run_sles12",
    "Product/SLES11-SP3" => "/usr/local/bin/script2run_sles11",
    "Product/SLES11-SP4" => "/usr/local/bin/script2run_sles11",
};
```

Regular expressions or substrings can be used to define a script for more than one repository in one project. The use of regular expressions needs to be activated by defining `$unpublishedhook_use_regex = 1;` (in `/usr/lib/obs/server/BSConfig.pm`):

```
our $unpublishedhook_use_regex = 1;
our $unpublishedhook = {
    "Product\/SLES12"     => "/usr/local/bin/script2run_sles12",
    "Product\/SLES11.*"   => "/usr/local/bin/script2run_sles11",
};
```

With custom parameters:

```
our $unpublishedhook_use_regex = 1;
our $unpublishedhook = {
    "Product\/SLES11.*" => [
      "/usr/local/bin/script2run", "sles11", "/srv/www/public_mirror"
    ],
};
```

The configuration is read by the publisher at startup only, so it has to be restarted after configuration changes have been made. The hook script's output is not logged by the publisher and should be written to a log file by the script itself. In case of a broken script, this is logged in the publisher's log file (/srv/obs/log/publisher.log by default):

```
Mon Mar  7 14:34:17 2016 publishing Product/SLES12
    fetched 0 patterns
    running createrepo
    calling unpublished hook /usr/local/bin/script2run_sles12
    /usr/local/bin/script2run_sles12 failed: 65280
    syncing database (6 ops)
```

Interactive scripts are not working and will fail immediately.

If you need to do a lot of work in the hook script and do not want to block the publisher all the time, consider using a separate daemon that does all the work and just gets triggered by the configured hook script.

The scripts are called without a timeout.



Note

Reminder: If *unpublish hooks* and *publish hooks* are defined, the *unpublish hook* runs before the *publish hook*.

5.7.2 Example Unpublisher Scripts

5.7.2.1 Simple Unpublisher Hook

The following example script deletes all packages from the target directory that have been removed from the repository.

```
#!/bin/bash
OBSHOME="/srv/obs"
LOGFILE="$OBSHOME/log/reposync.log"
DST_REPO_DIR="/srv/repo-mirror"
# Global substitution! To handle strings like Foo:Bar:testing - two
#+double-colons!
PRJ_PATH=${1//:/:\/}
PATH_TO_REPO=$2

shift 2
```

```
while [ $# -gt 0 ]
do
  rm -v $DST_REPO_DIR/$PRJ_PATH/$1 >>$LOGFILE 2>&1
  shift
done
```

For testing purposes, it can be invoked as follows:

```
$ sudo -u obsrun /usr/local/bin/unpublish-hook.sh \
    Product/SLES11-SP1                            \
    /srv/obs/repos/Product/SLE11-SP1              \
    src/icinga-1.13.3-1.3.src.rpm                 \
    x86_64/icinga-1.13.3-1.3.x86_64.rpm           \
    x86_64/icinga-devel-1.13.3-1.3.x86_64.rpm
```

5.7.2.2 Advanced Unpublisher Hook

The following example script reads the destination path from a parameter that is configured via the hook script:

```
#!/bin/bash
OBSHOME="/srv/obs"
LOGFILE="$OBSHOME/log/reposync.log"
DST_REPO_DIR=$1
# Global substitution! To handle strings like Foo:Bar:testing - two
#+double-colons!
PRJ_PATH=${1//:/:\/}
PATH_TO_REPO=$2

shift 3

while [ $# -gt 0 ]
do
  rm -v $DST_REPO_DIR/$PRJ_PATH/$1 >>$LOGFILE 2>&1
  shift
done
```

For testing purposes, it can be invoked as follows:

```
$ sudo -u obsrun /usr/local/bin/unpublish-hook.sh \
    /srv/www/public_mirror/Product/SLES11-SP1     \
    /srv/obs/repos/Product/SLE11SP1               \
    src/icinga-1.13.3-1.3.src.rpm                 \
    x86_64/icinga-1.13.3-1.3.x86_64.rpm           \
```

```
    x86_64/icinga-devel-1.13.3-1.3.x86_64.rpm
```

5.8 Managing Users and Groups

The OBS has an integrated user and group management with a role based access rights model. In every OBS instance, at least one user need to exist and have the global Admin role assigned. Groups can be defined by the Admin and instead of adding a list of users to a project/package role user can be added to a group and the group will be added to a project or package role.

5.8.1 User and Group Roles

The OBS role model has one global role: Admin, which can be granted to users. An OBS admin has access to all projects and packages via the API interface and the web user interface. Some menus in the Web UI do not allow changes by an Admin (for example, the Repository menu) as long the Admin is not a Maintainer for the project as well. But the same change can be done via editing the metadata directly. The other roles are specific to projects and packages and can be assigned to a user or a group.

TABLE 5.2: ROLES IN OBS

Role	Description	Remarks
Maintainer	Read and write access to projects or packages	
Bugowner	Read access to projects or packages	should be unique per package
Reader	Read access to sources	
Downloader	Read access to the binaries	
Reviewer	Default reviewer for a package or project	

5.8.2 Standalone User and Group Database

OBS provides its own user database which can also store a password. The authentication to the API happens via HTTP BASIC AUTH. See the API documentation to find out how to create, modify or delete user data. Also a call for changing the password exists.

Users can be added by the maintainer or if registration is allowed via the registration menu on the Web UI. It can be configured that a confirmation is needed after registration before the user may login.

5.8.3 Users and Group Maintainers

Administrators can create groups, add users to them, remove users from them and give Maintainer rights to users. This way, a maintainer will be able to also add, remove and give maintainer rights to other users.

```
osc api -d "<group><title><group-title></title><email><group-email></email><maintainer
 userid="<user-name>"/><person><person userid="<user_name>"/></person></group>' -X PUT "/
group/<group-title>"
```

5.8.4 Gravatar for Groups

In certain cases, it might be desirable to show a Gravatar for a group, similar to the users. In order to show a Gravatar, an email address is needed. Therefore, it is necessary that an admin adds an email address to the group through the API. This can be a achieved by

```
osc api -X POST "/group/<group-title>?cmd=set_email&email=<groups-email-address>"
```

5.8.5 Proxy Mode

The proxy mode can be used for specially secured instances, where the OBS web server shall not get connected to the network directly. There are authentication proxy products out there which do the authentication and send the user name via an HTTP header to OBS. Originally, this was developed for IChain - a legacy single login authentication method from Novell. This also has the advantage that the user password never reaches OBS.

The proxy mode can also be used for LDAP or Active Directory, but only for authentication.



Important

With enabled proxy mode the OBS trust the username in the http header. Since this was verified by the Web server and the Web server only forward requests for a verified and authenticated session, this is safe, as long you make sure that the direct web/API interface of the OBS is not reachable from the outside.

With the proxy mode the user still need to be registered in the OBS and all OBS roles and user properties are managed inside the OBS.

5.8.5.1 OBS Proxy Mode Configuration

Currently the LDAP configuration is in the *options.yml* file.

TABLE 5.3: OPTIONS FOR PROXY MODE CONFIGURATION

Config item	Description	Values `default`	Remarks
proxy_auth_mode	turn proxy mode on/off	`:off` :on	need to be :off if ldap_mode: is :on

5.8.6 LDAP/Active Directory



Note

The LDAP support was considered experimental and not officially supported. It is officially supported since 2.8.3 release.

Using LDAP or Active Directory as source for user and optional group information in environments which already have such a server has the advantage for the admin people that the user related information only need to be maintained in one place. In the following sections we are writing LDAP, but this includes Microsoft's Active Directory as well. Only in parts where differences exists Active Directory (AD) will be explicit mentioned.

In this mode the OBS contact the LDAP server directly from the OBS API, if the user was found and provides the correct password the user is added transparently to the OBS user database. The password or password hash is not stored in the OBS database. Because the user database password field is mandatory, a random hash is stored instead. The LDAP interface allows to restrict the access to users which are in a special LDAP group. Optional also groups can be discovered from the LDAP server. This can be also filtered.

Before anybody can add a user to a package or project with a role, the user need to had logged in at least one time, since the check for available users is local only. If the LDAP group mode is enabled, LDAP groups are also added transparently, if an existing group on the LDAP server is added to a project or package.

On bigger installations this mode can result in many search requests to the LDAP server and slow down access to projects and packages, because on every role check an LDAP search operation will contact the LDAP server. As alternative method group mirroring was implemented. This allows that the internal OBS group database is updated with the group membership information during the user authentication. All role test are made local against the OBS database and do not need additional LDAPoperations.



Note

The local user group membership in :mirror mode is updated as follows: When the user logins, the user *memberOf* attributes are parsed and compared with the global OBS grouplist, if a group matches, the user is added, if they are no longer a group member, they are removed. since this maybe a costly operation, depending on the group counts, this is only done on a full login. After a full login the user status is cashed for 2 minutes, if the user do a login during this time, nothing will be checked or updated. Here is a second mechanism to update user membership: If somebody adds a new Group in the OBS, the *member* attributes of the group are parsed and all current users which are in the local database become members.

5.8.6.1 OBS LDAP Configuration

Currently the main OBS LDAP configuration is in the file `options.yml`. Beside the settings in that file, the openLDAP configuration file is also evaluated by the Ruby LDAP implementation. This configuration file is usually located at `/etc/openldap/ldap.conf`. You can set here additional TLS/SSL directives like `TLS_CACERT`, `TLS_CACERTDIR` and `TLS_REQCERT`. For more information refer to the openLDAP man page (**`man ldap.conf`**).



Note

When LDAP mode is activated, users can only log in via LDAP. This also includes existing admin accounts. To make a LDAP user an admin, use a rake task which can be run on the OBS instance. For example, to make user `tux`, use:

```
cd /srv/www/obs/api
bundle exec rake user:give_admin_rights tux RAILS_ENV=production
```

TABLE 5.4: LDAP CONFIGURATION OPTIONS

Config item	Description	Values `default`	Remarks
ldap_mode	OBS LDAP mode on/off	`:off` :on	
ldap_servers	List of LDAP servers		colon-separated list
ldap_max_attempts	tries to ping LDAP server	int `15`	
ldap_search_timeout	timeout of an LDAP search	int 0…N `5`	0 wait for ever
ldap_user_memberof_attr	User attribute for Group membership	`memberOf`	case sensitive
ldap_group_member_attr	Group attribute for members	`member`	
ldap_ssl	use ldaps port and protocol	`:off` :on	
ldap_start_tls	usr Start TLS on LDAP protocol	:off `:on`	
ldap_port	LDAP portnumbers		if not set 389 for LDAP, 636 for LDAPS
ldap_referrals	Windows 2003 AD requires	`:off` :on	
ldap_search_base	company's LDAP search base for the users who will use OBS	`none`	
ldap_search_attr	user ID attribute	`sAMAccountName` uid	sAMAccountName for AD, uid for openldap
ldap_name_attr	Full user name	`cn`	

Config item	Description	Values `default`	Remarks
ldap_mail_attr	Attribute for users email	`mail`	
ldap_search_user	Bind user for LDAP search		for example, cn=ldap-bind, ou=system, dc=my-company, dc=com
ldap_search_auth	Password for the ldap_search_user		
ldap_user_filter	Search filter for OBS users		for example, a group membership, empty all users allowed
ldap_authenticate	How user how the credentials are verified	`:ldap` :local	only use :ldap
ldap_auth_mech	Used auth mech	`:md5` :cleartext	only if local
ldap_auth_attr	Used auth attribute for :local	`userPassword`	do not use
ldap_group_support	Import OBS groups from LDAP	`:off` :on :mirror	see text
ldap_group_search_base	company's LDAP search base for groups		
ldap_group_title_attr	Attribute of the group name	`cn`	
ldap_group_objectclass_attr	Object class for group	`Group`	
ldap_obs_admin_group	Group name for OBS Admins		if set, members of that group become OBS admin role

Example LDAP section of the *options.yml* file:

```
[...]
##################
# LDAP options
##################

ldap_mode: :on
# LDAP Servers separated by ':'.
# OVERRIDE with your company's ldap servers. Servers are picked randomly for
# each connection to distribute load.
ldap_servers: ldap1.mycompany.com:ldap2.mycompany.com

# Max number of times to attempt to contact the LDAP servers
ldap_max_attempts: 15

# timeout of an ldap search requests to avoid infinitely lookups (in seconds, 0 no
 timeout)
ldap_search_timeout: 5

# The attribute the user member of is stored in (case sensitive !)
ldap_user_memberof_attr: memberOf

# Perform the group_user search with the member attribute of group entry or memberof
 attribute of user entry
# It depends on your ldap define
# The attribute the group member is stored in
ldap_group_member_attr: member

# If you're using ldap_authenticate=:ldap then you should ensure that
# ldaps is used to transfer the credentials over SSL or use the StartTLS extension
ldap_ssl: :on

# Use StartTLS extension of LDAP
ldap_start_tls: :off

# LDAP port defaults to 636 for ldaps and 389 for ldap and ldap with StartTLS
#ldap_port:
# Authentication with Windows 2003 AD requires
ldap_referrals: :off

# OVERRIDE with your company's ldap search base for the users who will use OBS
ldap_search_base: ou=developmentt,dc=mycompany,dc=com
# Account name attribute (sAMAccountName for Active Directory, uid for openLDAP)
ldap_search_attr: sAMAccountName
# The attribute the users name is stored in
ldap_name_attr: cn
```

```
# The attribute the users email is stored in
ldap_mail_attr: mail
# Credentials to use to search ldap for the username
ldap_search_user: "cn=ldapbind,ou=system,dc=mycompany,dc=com"
ldap_search_auth: "top secret"

# By default any LDAP user can be used to authenticate to the OBS
# In some deployments this may be too broad and certain criteria should
# be met; eg group membership
#
# To allow only users in a specific group uncomment this line:
ldap_user_filter: (memberof=cn=obsusers,ou=groups,dc=mycompany,dc=com)
#
# Note this is joined to the normal selection like so:
# (&(#{dap_search_attr}=#{login})#{ldap_user_filter})
# giving an ldap search of:
#  (&(sAMAccountName=#{login})(memberof=CN=group,OU=Groups,DC=Domain Component))
#
# Also note that openLDAP must be configured to use the memberOf overlay

# ldap_authenticate says how the credentials are verified:
#   :ldap = attempt to bind to ldap as user using supplied credentials
#   :local = compare the credentials supplied with those in
#            LDAP using #{ldap_auth_attr} & #{ldap_auth_mech}
#       if :local is used then ldap_auth_mech can be
#       :md5
#       :cleartext
ldap_authenticate: :ldap
ldap_auth_mech: :md5
# This is a string
ldap_auth_attr: userPassword

# Whether to search group info from ldap, it does not take effect it is not set
# Please also set below ldap_group_* configs correctly to ensure the operation works
 properly
# Possible values:
#         :off     disabled
#         :on      enabled; every group member operation ask the LDAP server
#         :mirror  enabled; group membership is mirrored and updated on user login
#
ldap_group_support: :mirror

# OVERRIDE with your company's ldap search base for groups
ldap_group_search_base: ou=obsgroups,dc=mycompany,dc=com

# The attribute the group name is stored in
ldap_group_title_attr: cn
```

```
# The value of the group objectclass attribute
# group for Active Directory, groupOfNames in openLDAP
ldap_group_objectclass_attr: group

# The LDAP group for obs admins
# if this group is set and a user belongs to this group they get the global admin role
#
ldap_obs_admin_group: obsadmins
```

5.8.7 Authentication Methods

5.8.7.1 LDAP Methods

The LDAP mode has 2 methods to check authorization:

1. LDAP bind method.
   With the provided credentials, an LDAP bind request is tried.
2. Local method.
   The provided credentials checked locally against the content of the *userPassword* attribute.



Important

The local method should be not used, since the *userPassword* attribute in most LDAP installations will not be available until you are bind with a privilege user.

5.8.7.2 Kerberos

In OBS you can use single sign on via Kerberos tickets.

OBS Kerberos configuration resides in the *options.yml* file.

TABLE 5.5: KERBEROS CONFIGURATION OPTIONS

Config item	Description	Example
kerberos_keytab	Kerberos key table: file where long-term keys for one or more principals are stored	"/etc/krb5.keytab"

Config item	Description	Example
kerberos_service_principal	Kerberos OBS principal: OBS unique identity to which Kerberos can assign tickets	"HTTP/hostname.example.com@EXAMPLE.COM"
kerberos_realm	Kerberos realm: authentication administrative domain	"EXAMPLE.COM"

Example Kerberos section of the `options.yml` file:

```
[...]

##################
# Kerberos options
##################

kerberos_mode: true
kerberos_keytab: "/etc/krb5.keytab"
kerberos_service_principal: "HTTP/hostname.example.com@EXAMPLE.COM"
kerberos_realm: "EXAMPLE.COM"

[...]
```



Note

Once Kerberos is enabled, only users with logins that match users known to Kerberos will be able to authenticate to OBS. It is recommended to give admin rights to a matching user before enabling Kerberos mode.

5.8.7.3 OBS Token Authorization

OBS 2.5 provides a mechanism to create tokens for specific operations. This can be used to allow certain operations in the name of a user to others. This is esp. useful when integrating external infrastructure. The create token should be kept secret by default, but it can also be revoked at any time if it became obsolete or leaked.

5.8.7.3.1 Managing Tokens of a User

Tokens belong always to a user. A list of active tokens can received via

```
osc token
```

```
osc token --delete <TOKEN>
```

5.8.7.3.2 Executing a Source Service

A token can be used to execute a source service. The source service has to be setup for the package first, check the source service chapter for this. A typical example is to update sources of a package from git. A source service for that can be setup with

```
osc add git://....
```

A token can be registered as generic token, means allowing to execute all source services in OBS if the user has permissions. You can create such a token and execute the operation with

```
osc token --create
```

```
osc token --trigger <TOKEN> <PROJECT> <PACKAGE>
```

```
osc api -X POST /trigger/runservice?token=<TOKEN>&project=<PROJECT>&package=<PACKAGE>
```

You can also limit the token to a specific package. The advantage is that the operation is limited to that package, so less bad things can happen when the token leaks. Also you do not need to specify the package on execution time. Create and execute it with

```
osc token --create <PROJECT> <PACKAGE>
```

```
osc token --trigger <TOKEN>
```

```
osc api -X POST /trigger/runservice?token=<TOKEN>
```

5.9 Message Bus for Event Notifications

The OBS has an integrated notification subsystem for sending events that are happening in our app through a message bus. We have chosen RabbitMQ (https://www.rabbitmq.com/) as our message bus server technology based on the AMQP (https://www.amqp.org/) protocol.

5.9.1 RabbitMQ

RabbitMQ claims to be *"the most popular open source message broker"*. Meaning that it can deliver asynchronous messages in many different exchange ways (one to one, broadcasting, based on topics). It also includes a flexible routing system based on queues.

RabbitMQ is lightweight and easy to deploy on premises and in the cloud. It supports multiple messaging protocols too. And can be deployed in distributed and federated configurations to meet high-scale, high-availability requirements.

5.9.1.1 Configuration

Currently the RabbitMQ configuration is in the file `options.yml`. All those options there start with the prefix *amqp*. These configuration items match with some of the calls we do using the Bunny (http://rubybunny.info/) gem.

TABLE 5.6: RABBITMQ CONFIGURATION OPTIONS

Config item	**Description**	**Values** `default`	**Remarks**
amqp_namespace	Namespace for the queues of this instance	`'opensuse.obs'`	Is a prefix for the queue names
amqp_options	Connection configuration		See this guide (http://rubybunny.info/articles/connecting.html) to know which are the parameters allowed.
amqp_options[host]	Server host		A valid hostname
amqp_options[port]	Server port	`5672`	
amqp_options[user]	User account		
amqp_options[pass]	Account password		
amqp_options[vhost]	Virtual host		
amqp_exchange_name	Name for the exchange		

Config item	Description	Values `default`	Remarks
amqp_exchange_options	Exchange configuration		See this guide (http://rubybunny.info/articles/exchanges.html) to know more about exchanges.
amqp_exchange_options[type]	Type of communication for the exchange	`direct`	
amqp_exchange_options[auto_delete]	If set, the exchange is deleted when all queues have finished using it	`false`	
amqp_exchange_options[arguments]	More configuration for plugins / extensions		
amqp_queue_options	Queues configuration		See this guide (http://rubybunny.info/articles/queues.html) to know more about queues.
amqp_queue_options[durable]	Should this queue be durable?	`false`	
amqp_queue_options[auto_delete]	Should this queue be automatically deleted when the last consumer disconnects?	`false`	
amqp_queue_options[exclusive]	Should this queue be exclusive (only can be used by this connection, removed when the connection is closed)?	`false`	

Config item	Description	Values default	Remarks
amqp_queue_options[arguments]	Additional optional arguments (typically used by RabbitMQ extensions and plugins)		

Example of the RabbitMQ section of the *options.yml* file:

```
[...]
# RabbitMQ based message bus
#
# Prefix of the message bus rooting key

amqp_namespace: 'opensuse.obs'

# Connection options -> http://rubybunny.info/articles/connecting.html

amqp_options:
  host: rabbit.example.com
  port: 5672
  user: guest
  pass: guest
  vhost: /vhost

# Exchange options -> http://rubybunny.info/articles/exchanges.html

amqp_exchange_name: pubsub
amqp_exchange_options:
  type: topic
  auto_delete: false
  arguments:
    persistent: true
    passive: true

# Queue options -> http://rubybunny.info/articles/queues.html
amqp_queue_options:
  durable: false
  auto-delete: false
  exclusive: false
  arguments:
    extension_1: blah
```

TABLE 5.7: LIST OF EVENT MESSAGES / QUEUES FOR THE MESSAGE BUS

Queue Name	Description	Payload
__*prefix*__.package.build_success	A package build has succeeded	:project, :package, :repository, :arch, :release, :readytime, :srcmd5, :rev, :reason, :bcnt, :verifymd5, :starttime, :endtime, :workerid, :versrel, :buildtype
__*prefix*__.package.build_fail	A package build has failed	:project, :package, :repository, :arch, :release, :readytime, :srcmd5, :rev, :reason, :bcnt, :verifymd5, :starttime, :endtime, :workerid, :versrel, :previouslyfailed, :successive_failcount, :buildtype
__*prefix*__.package.build_unchanged	A package build has succeeded with unchanged result	:project, :package, :repository, :arch, :release, :readytime, :srcmd5, :rev, :reason, :bcnt, :verifymd5, :start-

Queue Name	Description	Payload
		time, :endtime, :workerid, :versrel, :buildtype
__*prefix*__.package.create	A new package was created	:project, :package, :sender
__*prefix*__.package.update	The package metadata was updated	:project, :package, :sender
__*prefix*__.package.delete	A package was deleted	:project, :package, :sender, :comment
__*prefix*__.package.undelete	A package was undeleted	:project, :package, :sender, :comment
__*prefix*__.package.branch	A package was branched	:project, :package, :sender, :targetproject, :targetpackage, :user
__*prefix*__.package.commit	A package has committed changes	:project, :package, :sender, :comment, :user, :files, :rev, :requestid
__*prefix*__.package.upload	Sources of a package were uploaded	:project, :package, :sender, :comment, :filename, :requestid, :target, :user

Queue Name	Description	Payload
__*prefix*__.package.service_success	Source service succeeded for a package	:comment, :project, :package, :sender, :rev, :user, :requestid
__*prefix*__.package.service_fail	Source service failed for a package	:comment, :error, :project, :package, :sender, :rev, :user, :requestid
__*prefix*__.package.version_change	A package has changed its version	:project, :package, :sender, :comment, :requestid, :files, :rev, :newversion, :user, :oldversion
__*prefix*__.package.comment	A new comment for the package was created	:project, :package, :sender, :commenters, :commenter, :comment_body, :comment_title
__*prefix*__.project.create	A new project was created	:project, :sender
__*prefix*__.project.update_project_conf	The project configuration was updated	:project, :sender, :files, :comment
__*prefix*__.project.update	A project was updated	:project, :sender
__*prefix*__.project.delete	A project was deleted	:project, :comment, :requestid, :sender
__*prefix*__.project.undelete	A project was undeleted	:project, :comment, :sender

Queue Name	Description	Payload
__*prefix*__.project.comment	A new comment for the project was created	:project, :commenters, :commenter, :comment_body, :comment_title
__*prefix*__.repo.packtrack	Binary was published in the repository	:project, :repo, :payload
__*prefix*__.repo.publish_state	Publish State of Repository has changed	:project, :repo, :state
__*prefix*__.repo.published	A repository was published	:project, :repo
__*prefix*__.repo.build_started	Repository (re)started building	:project, :repo, :arch, :buildid
__*prefix*__.repo.build_finished	Repository finished building	:project, :repo, :arch, :buildid
__*prefix*__.repo.status_report	Status Check for Finished Repository Created	:project, :repo, :arch, :buildid
__*prefix*__.request.create	A request was created	:author, :comment, :description, :number, :actions, :state, :when, :who, :diff (local projects)
__*prefix*__.request.change	A request was changed (admin only)	:author, :comment, :description, :number, :actions, :state, :when, :who
__*prefix*__.request.delete	A request was deleted	:author, :comment, :description, :number, :actions, :state, :when, :who

Queue Name	Description	Payload
__*prefix*__.request.state_change	The state of a request was changed	:author, :comment, :description, :number, :actions, :state, :when, :who, :oldstate
__*prefix*__.request.review_wanted	A request requires a review	:author, :comment, :description, :number, :actions, :state, :when, :who, :reviewers, :by_user, :by_group, :by_project, :package, :diff (local projects)
__*prefix*__.request.review_changed	Request was reviewed	:reviewers, :by_user, :by_group, :by_project, :package
__*prefix*__.request.reviews_done	All reviews of request have been completed	:author, :comment, :description, :number, :actions, :state, :when, :who, :reviewers, :by_user, :by_group, :by_project, :package, :diff (local projects)
__*prefix*__.request.comment	A new comment for the request was created	:id, :author, :comment, :description, :number, :actions, :state, :when, :who, :commenters, :commenter, :comment_body, :commen-

Queue Name	Description	Payload
		t_title, :request_number, :number, :diff_ref, :diff_file_index, :diff_line_number
__*prefix*__.request.status_report		:number
__*prefix*__.published.status_report	Status Check for Published Repository Created	:project, :repo, :buildid
__*prefix*__.container.published	A container has been published	:project, :repo, :buildid, :container
__*prefix*__.relationship.create	A user or group role has been added	:who, :user, :group, :project, :package, :role, :notifiable_id
__*prefix*__.relationship.delete	A user or group role has been removed	:who, :user, :group, :project, :package, :role, :notifiable_id

5.10 Backup

Open Build Service configuration and content needs usually a backup. The following explains suggested strategies and places considered for a backup.

5.10.1 Places to consider

The following is pointing to the places with admin configurations or user content. The default location places are considered here.

5.10.1.1 Frontend Configuration

- /srv/www/obs/api/config
- /srv/www/obs/api/log (optional)

The configuration is not changing usually. It is enough to backup it after config changes.

5.10.1.2 Frontend Database

The MySQL/MariaDB database backup can be done in different ways. Please consider the database manual for details. One possible way is to create dumps via mysqldump tool. The backup should be done at the same point of time as the source server. Inconsistencies can be resolved using the check_consistency tool.

5.10.1.3 Backend Configuration

The backend has a single configuration file which may got altered. This is by default /usr/lib/obs/server/BSConfig.pm . The file is not supposed to be changed usually and it can only be done by the system root user. A backup after a change is sufficient.

5.10.1.4 Backend Content

All backend content is below /srv/obs directory. This include the sources, build results and also all configuration changes done by the OBS admin users.

5.10.2 Backup strategies

A backup is ideally taken only from a not running service. In real live this is usually not possible, so it is important to run a backup on a production system.

5.10.2.1 Database

MySQL backup either directly from a non-primary node in the galera cluster (table dump locks the database during operation) or from a mysql slave attached to the cluster.

5.10.2.2 Sources

The sources are supposed to be backup at the same time as the database. This can get achieved by either having a dedicated instance for the source server or by having a backup of the following directories.

- /srv/obs/projects
- /srv/obs/sources

5.10.2.3 Build Results

Full backups via snapshots, either offered by the SAN storage or via LVM snapshot methods. Consistency is normally on repository level. Any inconsistency will be found by the scheduler and content will be retriggered. This is not true for disabled builds like released builds.

5.11 Restore

A restored system might contain inconsistencies if it was taken from a running service. These can be resolved as follows.

5.11.1 Check and repair database inconsistencies

If either database portions or sources got restored there are chances for inconsistencies. These can be found via

```
geeko > cd /srv/www/obs/api/
geeko > ./bin/rails c
geeko > ConsistencyCheckJob.new.perform
```

Single projects can be either checked with

```
geeko > cd /srv/www/obs/api/
geeko > ./bin/rake check_project project="YOUR_PROJECT"
```

or inconsistencies fixed via

```
geeko > cd /srv/www/obs/api/
geeko > ./bin/rake fix_project project="YOUR_PROJECT"
```

5.11.2 Binaries

All build results are evaluated by the scheduler. Therefore any inconsistency can be detected by the scheduler. One way is to enforce a cold start, which means that the scheduler would rescan all sources and binaries and trigger builds where needed. This can be achieved by

```
geeko > rcobsscheduler stop     # ensure no scheduler is running
geeko > rm /srv/obs/run/*.state # remove all state files
geeko > rcobsscheduler start
```

The scheduler state will be visible as in cold start. It may take a longer time, so it might be more efficient to check only certain projects or architectures if needed. This can be triggered in a running system by executing

```
geeko > obs_admin --check-project PROJECT ARCHITERCTURE
```

A deep check is necessary in case sources have been restored:

```
geeko > obs_admin --deep-check-project PROJECT ARCHITERCTURE
```

5.12 Repair Data Corruption

On-disk data might be corrupted independent of a restore. For example due to power outage, filesystem or disk errors. A MySQL/Maria database in a cluster should repair itself in that case. Data on disk in the backend parts can be checked and fixed using an dedicated tool. See the help of the tool for further details or run

```
geeko > /usr/lib/obs/server/bs_check_consistency --check-all
```

Data can be repaired using the fix options.

5.13 Spider Identification

OBS is hiding specific parts/pages of the application from search crawlers (DuckDuckGo, Google, etc.), mostly for performance reasons. Which user-agent strings are identified as crawlers configured in the file `/srv/www/obs/api/config/crawler-user-agents.json`.

To update that list, you must run the command **`bundle exec rake voight_kampf:import_user_agents`** in the root directory of your OBS instance. This downloads the current crawler list of user agents as a JSON file into the `config/` directory of the Rails application.

If you want to extend or edit this list, switch to the `config/` directory and open the `crawler-user-agents.json` file with the editor of your choice. The content can look like this:

```
[
   {
       "pattern": "Googlebot\\/",
       "url": "http://www.google.com/bot.html"
   },
   {
       "pattern": "Googlebot-Mobile"
   },
   {
       "pattern": "Googlebot-Image"
   },
   [...]
]
```

To add a new bot to this list, a pattern must be defined. This is required to identify a bot. Almost all bots have their own user agent that they are sending to a Web server to identify them. For example, the user agent of the Googlebot looks like this:

```
Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)
```

To choose the pattern for the new bot, compare the user agent of the bot you want to identify with others and look for a part that is unique (like in the Googlebot example, the part: Googlebot).

Let's assume we want to add the bot Geekobot to the list of bots and the user agent looks like this:

```
Mozilla/5.0 (compatible; Geekobot/2.1; +https://www.opensuse.org)
```

Our unique part would be Geekobot. So we add a new entry to the list of bots:

```
[
   {
       "pattern": "Googlebot\\/",
       "url": "http://www.google.com/bot.html"
   },
   {
       "pattern": "Googlebot-Mobile"
   },
   {
       "pattern": "Googlebot-Image"
   },
   [...]
   {
```

```
        "pattern": "Geekobot"
    }
]
```



Note

You can also use regular expressions in the pattern element.

Save the file and restart the Rails application and the bot Geekobot should be identified properly.

5.14 Worker in Kubernetes



Warning: Alpha Implementation

This is Alpha implementation and not recommended for production.

The Kubernetes device plugin deployed here makes several assumptions about which and how many containers will have access to KVM device.

The plugin also assumes availability of /dev/kvm on every node where the device-plugin-container is running

The build service worker itself has many backends to run its jobs. One of the preferred backends is KVM.

This backend allows building inside a VM. This has many advantages from security and isolation perspective.

When a build worker is running inside the containerized environment (for example, using Kubernetes) access to KVM is not available.

For such situations Kubernetes provides access to host devices (for example: KVM, GPU…) through device plugins.

So, `/dev/kvm` can be made available to containers via Kubernetes using device plugin API (https://kubernetes.io/docs/concepts/extend-kubernetes/compute-storage-net/device-plugins/).

One of the implementations of K8s devices plugin for KVM is available here : https://github.com/kubevirt/kubernetes-device-plugins

1. Use the following manifest to deploy the KVM device plugin in a container.

This plugin is packaged as `k8s-device-plugin-kvm` and corresponding container built here: https://build.opensuse.org/package/show/home:sjamgade:branches:devel:CaaSP:Head:ControllerNode/kubernetes-device-plugins-docker

```
apiVersion: apps/v1
kind: Deployment
metadata:
   name: kvm-deployment
spec:
   replicas: 1
   selector:
     matchLabels:
        app: kvm-server
   template:
     metadata:
       labels:
         app: kvm-server
     spec:
       containers:
       - name: kvm-pod
         command: ["/usr/bin/k8s-kvm"]
         args: ["-v", "3","-logtostderr"]
         image: registry.opensuse.org/home/sjamgade/branches/devel/caasp/head/
controllernode/containers/my_container
         imagePullPolicy: IfNotPresent
         securityContext:
          capabilities:
            add:
            - NET_ADMIN
            - SYS_NICE
          privileged: True
          runAsUser: 0
         volumeMounts:
              - name: device-plugins-socket
                mountPath: /var/lib/kubelet/device-plugins
       hostname: kvm-server
       volumes:
         - name: device-plugins-socket
           hostPath:
           path: /var/lib/kubelet/device-plugins
```

2. Build container image of the build service locally and load it to all worker nodes. There is sample project file here: https://build.opensuse.org/package/show/home:sjamgade:branches:OBS:Server:Unstable/OBS-Appliance `docker load < "/path/to/docker.archive.tar.gz"`

3. Use the following manifest to deploy the build service worker.

   Here ports are hard-coded to allow easy integration with local kubelet without requiring a separate ingress-controller

```
apiVersion: apps/v1
kind: Deployment
metadata:
   name: worker-deployment-1
spec:
   replicas: 1
   selector:
     matchLabels:
        app: obsworkerappname
   template:
     metadata:
       labels:
         app: obsworkerappname
     spec:
       containers:
       - name: test-worker-pod
         command: ["/bin/bash"]
         args: ["-c", "sleep 1d  && echo Sleep expired > /dev/termination-log"]
         image: docker.io/library/obsworker
         imagePullPolicy: Never
         resources:
          limits:
            devices.kubevirt.io/kvm: "1"
          requests:
            cpu: 100m
            devices.kubevirt.io/kvm: "1"
         securityContext:
          capabilities:
            add:
            - NET_ADMIN
            - SYS_NICE
          privileged: false
          runAsUser: 0
         volumeMounts:
              - name: boot-dir
                mountPath: /boot
              - name: modules-dir
                mountPath: /lib/modules
       volumes:
         - name: boot-dir
           hostPath:
             path: /boot
```

```
          - name: modules-dir
            hostPath:
              path: /lib/modules
       hostname: obs-worker-1
---
apiVersion: v1
kind: Service
metadata:
  name: myobsservice
  labels:
    servicename: obsworkerservicename
spec:
  selector:
    app: obsworkerappname
  type: NodePort
  externalTrafficPolicy: "Local"
  ports:
    - name: woker-1
      protocol: TCP
      port: 32515
      targetPort: 32515
      nodePort: 32315
    - name: woker-2
      protocol: TCP
      port: 32516
      targetPort: 32516
      nodePort: 32516
```

4. Save the following into a file `launchworker.sh`. Later use this file to launch the worker. Make sure you uncomment the OBS_REPO_SERVERS line and change the IP address to your build servers address

```
cat << EOH > /etc/buildhost.config
OBS_WORKER_DIRECTORY=/var/cache/obs/worker
OBS_CACHE_DIR=/var/cache/obs/worker/cache
OBS_CACHE_SIZE=50140
#### CHANGE THIS TO YOUR SERVER ADDRESS
# OBS_REPO_SERVERS=192.168.132.113:5252
####
OBS_WORKER_INSTANCES=1
OBS_WORKER_PORTBASE=32516
OBS_WORKER_TEST_MODE=1
OBS_VM_TYPE=kvm
OBS_WORKER_WIPE_AFTER_BUILD=1
EOH
```

```
obsworker restart
```

5. Use the following command to launch the build service worker.

```
cat launchworker.sh | kubectl exec -i -t test-worker-pod bash
```

6 Troubleshooting

Here are two major classes of problems regarding the Open Build Service:

1. Normal package build errors
2. Bugs, resource shortage or config issues caused issues

The first category are errors like missing dependent packages in the build environment, errors during compiling or linking, errors in the build description and so on. Most of them should not happen if the packager does test the build locally before committing it to the OBS. This type of problems is not covered by this chapter.

6.1 General Hints

If you detect unexpected behavior of the OBS, you should follow some rules to locate the problem:

1. Consult the log files, for the back-end look at */srv/obs/log* for the back-end log files and */srv/www/obs/api/log* for the front-end log files. See the Log files *Section 3.2, "Log Files"* for more details.
2. Consult the normal OS system logs and the kernel log (dmesg) if here are reported system or HW problems.
3. Check if all services are running on the back-end and front-end. See the OBS Architecture in reference book for details.
4. Try to find an easy way to reproduce the problem.
5. To check whether this issue was already reported, see https://github.com/openSUSE/open-build-service.
6. Use search machines (Google) to find out if others did also run into this problem. If you are lucky, you will find a fix or workaround as well.
7. If you create a new bug report, include all information to reproduce the problem and the complete error message/error log if here are any.

6.2 Debugging Front-end Problems

If you get unexpected results from submitting commands with the ***osc*** tool, you can use the debug feature of the tools to find more information about what happened.

osc debug options

```
--debugger          jump into the debugger before executing anything
--post-mortem       jump into the debugger in case of errors
-t, --traceback     print call trace in case of errors
-H, --http-debug    debug HTTP traffic (filters some headers)
--http-full-debug   debug HTTP traffic (filters no headers)
-d, --debug         print info useful for debugging
```

The ***--debugger*** and ***--post-mortem*** are only suitable for *osc* developers. If you get an error message from osc, the ***-t, --traceback*** can give the developer some more information about the problem. The ***-H, --http-debug*** and ***--http-full-debug*** options are useful to see the raw answers of OBS API, often this gives a hint what maybe wrong. If you report a problem regarding the ***osc*** tool, it may help to include the osc output with **additional *--*http-debug *--traceback*** options.



Warning

With ***--http-full-debug*** all http headers are included, this may include user data and authentication stuff so review and replace such data with XXXXXXXX or so before you post it on the internet.

7 Setting Up a Local OBS Instance

This chapter explains how you could setup/Install/test OBS in your system. This chapter is written for those who are not so familiar with Linux and OBS. So in case you are confident to set up OBS, skip this chapter. Following would be explained in this chapter.

- OBS 1 click install, then manual setup in openSUSE 13.1;
- OBS 1 click install, then manual setup in SLES11;
- OBS test run on Microsoft Windows using VMware player;
- OBS appliance installed manually in a VirtualBox.

Unfortunately, didn't have a chance to install OBS in other Linux distribution yet. The last section will explain your first steps with the new OBS server.

How you could install and purchase openSUSE 13.1 and SLES12 will not be explained in this chapter. VMware player install and purchase also will not be explained. For these topics, you could visit for help:

- http://software.opensuse.org/131/en
- http://www.suse.com/products/server/
- http://my.vmware.com/web/vmware/free#desktop_end_user_computing/vmware_player/6_0

7.1 Testing OBS on Microsoft Windows Using VMware Player

Those who are not familiar with Linux can run and test OBS. To run and test OBS in Windows, you could use a virtual machine program such as VMware or VirtualBox, etc. This chapter explains, how you can run OBS using VMware player. To check and test with VirtualBox or another virtual machine, check in below.

After you are done installing VMware player on Windows, you need to download the OBS appliance program. You could get OBS appliance file by visiting http://openbuildservice.org/download/other/ and clicking on *VirtualBox/VMware Image*. After downloading, uncompress with some Windows archiving program that understands the `.tar.bz2` file format.

Now, open VMware Player application and select *File* › *Open a Virtual Machine*, or you could press `Ctrl` – `O` directly. Open the decompressed virtual machine in *Open Virtual Machine* dialog box. Click on *Play virtual machine* icon or hyper link in VMware player.

At the Linux prompt, you can login using "root" as a login name and "opensuse" as a password. Now, OBS local instance should be already loaded and running in your system. To make sure that the OBS Web UI is successfully up and running, open the OBS Web UI.

To access OBS Web user interface, open your web browser and try the address `http://vm.ip.address`. You can check the virtual machine's IP address by using **`ifconfig`** Linux command. Now, you probably could see a screen like the one below in your window:

To login your local OBS instance, you could use default login name as "Admin" and password as "opensuse". Check if you could login properly by clicking *Login* on your local OBS instance Web UI.

7.2 Installing a Readymade OBS Appliance in a VirtualBox

This method is slightly less easy than the method using the readymade vmdk VMware disk, but it enables you to determine the size of your virtual disks to your convenience. It could also work with a real computer with two disks. It requires some knowledge of command line and partitioning.

1. Download the OBS appliance installer. Visit: http://www.openbuildservice.org/download and press the *Download the OBS Appliance Installer* button. It will start downloading an ISO image.
2. In VirtualBox, create a virtual machine with:
    - 4 GB memory
    - 1 virtual hard disk of 20 G for `/` and `/var/cache/obs`
    - 1 virtual hard disk of 50 G for `/srv/obs`
    - a virtual CD-ROM driver pointing to the downloaded ISO image
    - network bridging with real Ethernet card
3. Boot the virtual machine and choose to install the OBS server on the smaller virtual hard disk.
4. Log into the virtual machine with Login: root and Password: opensuse. If needed, switch to German/French/whatever keyboard: **`# loadkeys de`**. Inspect partitioning: **`# df -h`**. It shows you that the root partition is small and already almost full (1.6 GB used out of 1.8). Let's prepare the

other partitions to get a bit more working space. First, **`# fdisk /dev/sda`** and prepare `/dev/sda2` to use the remaining space. Second, **`# fdisk /dev/sdb`** and prepare `/dev/sdb1` to use all the space, with type 8e (Linux LVM):

```
# pvcreate /dev/sdb1
# vgcreate OBS /dev/sdb1
# lvcreate -n server -L 48G OBS
# mkfs.ext4 /dev/OBS/server
```

5. Reboot, this time onto the hard disk. The CD-ROM might be disconnected, we will not need it anymore. Log in as root user, change keyboard if needed, and format `/dev/sda2`: **`# mkfs.ext4 /dev/sda2`**. Add following entry to /etc/fstab:

   ```
   /dev/sda2  /var/cache/obs  ext4  defaults  2 1
   ```

   Mount the new partition: **`# mount /dev/sda2`**. Get your IP address: **`# ifconfig`**.

7.3 First Steps with Your New OBS Server

At this point, one of the methods above should have provided you with a running OBS instance. Let us get our first package building.

1. From a web browser, access the web interface: https://vm.ip.address/. Accept the self-signed certificate.
2. In the top right corner of the web interface, there is a *Log In* option. Use it to log in as: Admin opensuse.
3. Click on the *Configuration* button to give your server a name and a description. Click on the *Interconnect* option. Choose *openSUSE* as the remote repository where we will pick up the packages of the build environment. Log out of the web interface.
4. Use *Sign Up* option to create a regular user account (for example: hmustermann).
5. As this normal user, click on the *Home Project* option and create your home project (that would be: "home:hmustermann").
6. Go to this home project, and click on *Create package* to create your first package (let's say: "mypackage").
7. Go back to your home project, and click the *Repositories* button. Choose to add a new repository and pick *openSUSE 13.1* (for example).

8. Reboot the virtual machine to ensure all projects are rescanned.

9. From outside the virtual machine and as a normal user, declare in `~/.oscrc` your new OBS user:

```
[https://vm.ip.address]
user=hmustermann
pass=bond007
```

   then checkout your new package: **`$ osc -A https://vm.ip.address co home:hmustermann`**. go to your first package: **`$ cd home:hmustermann/mypackage`**. and add some sources there (tarball, spec file, changelog, patches). Check them in, then trigger a remote build:

```
$ osc add *
$ osc commit
$ osc rebuild
```

10. The built packages can be seen at: http://vm.ip.address:82/

A GNU Licenses

This appendix contains the GNU General Public License version 2 and the GNU Free Documentation License version 1.2.

GNU General Public License

Version 2, June 1991

Copyright (C) 1989, 1991 Free Software Foundation, Inc. 59 Temple Place - Suite 330, Boston, MA 02111-1307, USA

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

Preamble

The licenses for most software are designed to take away your freedom to share and change it. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change free software--to make sure the software is free for all its users. This General Public License applies to most of the Free Software Foundation's software and to any other program whose authors commit to using it. (Some other Free Software Foundation software is covered by the GNU Library General Public License instead.) You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for this service if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs; and that you know you can do these things.

To protect your rights, we need to make restrictions that forbid anyone to deny you these rights or to ask you to surrender the rights. These restrictions translate to certain responsibilities for you if you distribute copies of the software, or if you modify it.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must give the recipients all the rights that you have. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

We protect your rights with two steps: (1) copyright the software, and (2) offer you this license which gives you legal permission to copy, distribute and/or modify the software.

Also, for each author's protection and ours, we want to make certain that everyone understands that there is no warranty for this free software. If the software is modified by someone else and passed on, we want its recipients to know that what they have is not the original, so that any problems introduced by others will not reflect on the original authors' reputations.

Finally, any free program is threatened constantly by software patents. We wish to avoid the danger that redistributors of a free program will individually obtain patent licenses, in effect making the program proprietary. To prevent this, we have made it clear that any patent must be licensed for everyone's free use or not licensed at all.

The precise terms and conditions for copying, distribution and modification follow.

GNU GENERAL PUBLIC LICENSE TERMS AND CONDITIONS FOR COPYING, DISTRIBUTION AND MODIFICATION

0. This License applies to any program or other work which contains a notice placed by the copyright holder saying it may be distributed under the terms of this General Public License. The "Program", below, refers to any such program or work, and a "work based on the Program" means either the Program or any derivative work under copyright law: that is to say, a work containing the Program or a portion of it, either verbatim or with modifications and/or translated into another language. (Hereinafter, translation is included without limitation in the term "modification".) Each licensee is addressed as "you".

Activities other than copying, distribution and modification are not covered by this License; they are outside its scope. The act of running the Program is not restricted, and the output from the Program is covered only if its contents constitute a work based on the Program (independent of having been made by running the Program). Whether that is true depends on what the Program does.

1. You may copy and distribute verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice and disclaimer of warranty; keep intact all the notices that refer to this License and to the absence of any warranty; and give any other recipients of the Program a copy of this License along with the Program.

You may charge a fee for the physical act of transferring a copy, and you may at your option offer warranty protection in exchange for a fee.

2. You may modify your copy or copies of the Program or any portion of it, thus forming a work based on the Program, and copy and distribute such modifications or work under the terms of Section 1 above, provided that you also meet all of these conditions:

a). You must cause the modified files to carry prominent notices stating that you changed the files and the date of any change.

b). You must cause any work that you distribute or publish, that in whole or in part contains or is derived from the Program or any part thereof, to be licensed as a whole at no charge to all third parties under the terms of this License.

c). If the modified program normally reads commands interactively when run, you must cause it, when started running for such interactive use in the most ordinary way, to print or display an announcement including an appropriate copyright notice and a notice that there is no warranty (or else, saying that you provide a warranty) and that users may redistribute the program under these conditions, and telling the user how to view a copy of this License. (Exception: if the Program itself is interactive but does not normally print such an announcement, your work based on the Program is not required to print an announcement.)

These requirements apply to the modified work as a whole. If identifiable sections of that work are not derived from the Program, and can be reasonably considered independent and separate works in themselves, then this License, and its terms, do not apply to those sections when you distribute them as separate works. But when you distribute the same sections as part of a whole which is a work based on the Program, the distribution of the whole must be on the terms of this License, whose permissions for other licensees extend to the entire whole, and thus to each and every part regardless of who wrote it.

Thus, it is not the intent of this section to claim rights or contest your rights to work written entirely by you; rather, the intent is to exercise the right to control the distribution of derivative or collective works based on the Program.

In addition, mere aggregation of another work not based on the Program with the Program (or with a work based on the Program) on a volume of a storage or distribution medium does not bring the other work under the scope of this License.

3. You may copy and distribute the Program (or a work based on it, under Section 2) in object code or executable form under the terms of Sections 1 and 2 above provided that you also do one of the following:

a). Accompany it with the complete corresponding machine-readable source code, which must be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,

b). Accompany it with a written offer, valid for at least three years, to give any third party, for a charge no more than your cost of physically performing source distribution, a complete machine-readable copy of the corresponding source code, to be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,

c). Accompany it with the information you received as to the offer to distribute corresponding source code. (This alternative is allowed only for noncommercial distribution and only if you received the program in object code or executable form with such an offer, in accord with Subsection b above.)

The source code for a work means the preferred form of the work for making modifications to it. For an executable work, complete source code means all the source code for all modules it contains, plus any associated interface definition files, plus the scripts used to control compilation and installation of the executable. However, as a special exception, the source code distributed need not include anything that is normally distributed (in either source or binary form) with the major components (compiler, kernel, and so on) of the operating system on which the executable runs, unless that component itself accompanies the executable.

If distribution of executable or object code is made by offering access to copy from a designated place, then offering equivalent access to copy the source code from the same place counts as distribution of the source code, even though third parties are not compelled to copy the source along with the object code.

4. You may not copy, modify, sublicense, or distribute the Program except as expressly provided under this License. Any attempt otherwise to copy, modify, sublicense or distribute the Program is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.

5. You are not required to accept this License, since you have not signed it. However, nothing else grants you permission to modify or distribute the Program or its derivative works. These actions are prohibited by law if you do not accept this License. Therefore, by modifying or distributing the Program (or any work based on the Program), you indicate your acceptance of this License to do so, and all its terms and conditions for copying, distributing or modifying the Program or works based on it.

6. Each time you redistribute the Program (or any work based on the Program), the recipient automatically receives a license from the original licensor to copy, distribute or modify the Program subject to these terms and conditions. You may not impose any further restrictions on the recipients' exercise of the rights granted herein. You are not responsible for enforcing compliance by third parties to this License.

7. If, as a consequence of a court judgment or allegation of patent infringement or for any other reason (not limited to patent issues), conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot distribute so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not distribute the Program at all. For example, if a patent license would not permit royalty-free redistribution of the Program by all those who receive copies directly or indirectly through you, then the only way you could satisfy both it and this License would be to refrain entirely from distribution of the Program.

If any portion of this section is held invalid or unenforceable under any particular circumstance, the balance of the section is intended to apply and the section as a whole is intended to apply in other circumstances.

It is not the purpose of this section to induce you to infringe any patents or other property right claims or to contest validity of any such claims; this section has the sole purpose of protecting the integrity of the free software distribution system, which is implemented by public license practices. Many people have made generous contributions to the wide range of software distributed through that system in reliance on consistent application of that system; it is up to the author/donor to decide if he or she is willing to distribute software through any other system and a licensee cannot impose that choice.

This section is intended to make thoroughly clear what is believed to be a consequence of the rest of this License.

8. If the distribution and/or use of the Program is restricted in certain countries either by patents or by copyrighted interfaces, the original copyright holder who places the Program under this License may add an explicit geographical distribution limitation excluding those countries, so that distribution is permitted only in or among countries not thus excluded. In such case, this License incorporates the limitation as if written in the body of this License.

9. The Free Software Foundation may publish revised and/or new versions of the General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies a version number of this License which applies to it and "any later version", you have the option of following the terms and conditions either of that version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of this License, you may choose any version ever published by the Free Software Foundation.

10. If you wish to incorporate parts of the Program into other free programs whose distribution conditions are different, write to the author to ask for permission. For software which is copyrighted by the Free Software Foundation, write to the Free Software Foundation; we sometimes make exceptions for this. Our decision will be guided by the two goals of preserving the free status of all derivatives of our free software and of promoting the sharing and reuse of software generally.

NO WARRANTY

11. BECAUSE THE PROGRAM IS LICENSED FREE OF CHARGE, THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

12. IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MAY MODIFY AND/OR REDISTRIBUTE THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAM-

AGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest possible use to the public, the best way to achieve this is to make it free software which everyone can redistribute and change under these terms.

To do so, attach the following notices to the program. It is safest to attach them to the start of each source file to most effectively convey the exclusion of warranty; and each file should have at least the “copyright” line and a pointer to where the full notice is found.

```
one line to give the program's name and an idea of what it does.
Copyright (C) yyyy name of author
```

```
This program is free software; you can redistribute it and/or
modify it under the terms of the GNU General Public License
as published by the Free Software Foundation; either version 2
of the License, or (at your option) any later version.
```

```
This program is distributed in the hope that it will be useful,
but WITHOUT ANY WARRANTY; without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
GNU General Public License for more details.
```

```
You should have received a copy of the GNU General Public License
along with this program; if not, write to the Free Software
Foundation, Inc., 59 Temple Place - Suite 330, Boston, MA 02111-1307, USA.
```

Also add information on how to contact you by electronic and paper mail.

If the program is interactive, make it output a short notice like this when it starts in an interactive mode:

```
Gnomovision version 69, Copyright (C) year name of author
Gnomovision comes with ABSOLUTELY NO WARRANTY; for details
type `show w'. This is free software, and you are welcome
to redistribute it under certain conditions; type `show c'
for details.
```

The hypothetical commands `show w’ and `show c’ should show the appropriate parts of the General Public License. Of course, the commands you use may be called something other than `show w’ and `show c’; they could even be mouse-clicks or menu items--whatever suits your program.

You should also get your employer (if you work as a programmer) or your school, if any, to sign a “copyright disclaimer” for the program, if necessary. Here is a sample; alter the names:

```
Yoyodyne, Inc., hereby disclaims all copyright
interest in the program `Gnomovision'
(which makes passes at compilers) written
by James Hacker.
```

```
signature of Ty Coon, 1 April 1989
Ty Coon, President of Vice
```

This General Public License does not permit incorporating your program into proprietary programs. If your program is a subroutine library, you may consider it more useful to permit linking proprietary applications with the library. If this is what you want to do, use the GNU Lesser General Public License (http://www.fsf.org/licenses/lgpl.html)↗ instead of this License.

GNU Free Documentation License

Version 1.2, November 2002

Copyright (C) 2000,2001,2002 Free Software Foundation, Inc. 59 Temple Place, Suite 330, Boston, MA 02111-1307 USA

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

PREAMBLE

The purpose of this License is to make a manual, textbook, or other functional and useful document “free” in the sense of freedom: to assure everyone the effective freedom to copy and redistribute it, with or without modifying it, either commercially or noncommercially. Secondarily, this License preserves for the author and publisher a way to get credit for their work, while not being considered responsible for modifications made by others.

This License is a kind of “copyleft”, which means that derivative works of the document must themselves be free in the same sense. It complements the GNU General Public License, which is a copyleft license designed for free software.

We have designed this License in order to use it for manuals for free software, because free software needs free documentation: a free program should come with manuals providing the same freedoms that the software does. But this License is not limited to software manuals; it can be used for any textual work, regardless of subject matter or whether it is published as a printed book. We recommend this License principally for works whose purpose is instruction or reference.

APPLICABILITY AND DEFINITIONS

This License applies to any manual or other work, in any medium, that contains a notice placed by the copyright holder saying it can be distributed under the terms of this License. Such a notice grants a world-wide, royalty-free license, unlimited in duration, to use that work under the conditions stated herein. The “Document”, below, refers to any such manual or work. Any member of the public is a licensee, and is addressed as “you”. You accept the license if you copy, modify or distribute the work in a way requiring permission under copyright law.

A “Modified Version” of the Document means any work containing the Document or a portion of it, either copied verbatim, or with modifications and/or translated into another language.

A “Secondary Section” is a named appendix or a front-matter section of the Document that deals exclusively with the relationship of the publishers or authors of the Document to the Document’s overall subject (or to related matters) and contains nothing that could fall directly within that overall subject. (Thus, if the Document is in part a textbook of mathematics, a Secondary Section may not explain any mathematics.) The relationship could be a matter of historical connection with the subject or with related matters, or of legal, commercial, philosophical, ethical or political position regarding them.

The “Invariant Sections” are certain Secondary Sections whose titles are designated, as being those of Invariant Sections, in the notice that says that the Document is released under this License. If a section does not fit the above definition of Secondary then it is not allowed to be designated as Invariant. The Document may contain zero Invariant Sections. If the Document does not identify any Invariant Sections then there are none.

The “Cover Texts” are certain short passages of text that are listed, as Front-Cover Texts or Back-Cover Texts, in the notice that says that the Document is released under this License. A Front-Cover Text may be at most 5 words, and a Back-Cover Text may be at most 25 words.

A “Transparent” copy of the Document means a machine-readable copy, represented in a format whose specification is available to the general public, that is suitable for revising the document straightforwardly with generic text editors or (for images composed of pixels) generic paint programs or (for drawings) some widely available drawing editor, and that is suitable for input to text formatters or for automatic translation to a variety of formats suitable for input to text formatters. A copy made in an otherwise Transparent file format whose markup, or absence of markup, has been arranged to thwart or discourage subsequent modification by readers is not Transparent. An image format is not Transparent if used for any substantial amount of text. A copy that is not “Transparent” is called “Opaque”.

Examples of suitable formats for Transparent copies include plain ASCII without markup, Texinfo input format, LaTeX input format, SGML or XML using a publicly available DTD, and standard-conforming simple HTML, PostScript or PDF designed for human modification. Examples of transparent image formats include PNG, XCF and JPG. Opaque formats include proprietary formats that can be read and edited only by proprietary word processors, SGML or XML for which the DTD and/or processing tools are not generally available, and the machine-generated HTML, PostScript or PDF produced by some word processors for output purposes only.

The “Title Page” means, for a printed book, the title page itself, plus such following pages as are needed to hold, legibly, the material this License requires to appear in the title page. For works in formats which do not have any title page as such, “Title Page” means the text near the most prominent appearance of the work’s title, preceding the beginning of the body of the text.

A section “Entitled XYZ” means a named subunit of the Document whose title either is precisely XYZ or contains XYZ in parentheses following text that translates XYZ in another language. (Here XYZ stands for a specific section name mentioned below, such as “Acknowledgements”, “Dedications”, “Endorsements”, or “History”.) To “Preserve the Title” of such a section when you modify the Document means that it remains a section “Entitled XYZ” according to this definition.

The Document may include Warranty Disclaimers next to the notice which states that this License applies to the Document. These Warranty Disclaimers are considered to be included by reference in this License, but only as regards disclaiming warranties: any other implication that these Warranty Disclaimers may have is void and has no effect on the meaning of this License.

VERBATIM COPYING

You may copy and distribute the Document in any medium, either commercially or noncommercially, provided that this License, the copyright notices, and the license notice saying this License applies to the Document are reproduced in all copies, and that you add no other conditions whatsoever to those of this License. You may not use technical measures to obstruct or control the reading or further copying of the copies you make or distribute. However, you may accept compensation in exchange for copies. If you distribute a large enough number of copies you must also follow the conditions in section 3.

You may also lend copies, under the same conditions stated above, and you may publicly display copies.

COPYING IN QUANTITY

If you publish printed copies (or copies in media that commonly have printed covers) of the Document, numbering more than 100, and the Document's license notice requires Cover Texts, you must enclose the copies in covers that carry, clearly and legibly, all these Cover Texts: Front-Cover Texts on the front cover, and Back-Cover Texts on the back cover. Both covers must also clearly and legibly identify you as the publisher of these copies. The front cover must present the full title with all words of the title equally prominent and visible. You may add other material on the covers in addition. Copying with changes limited to the covers, as long as they preserve the title of the Document and satisfy these conditions, can be treated as verbatim copying in other respects.

If the required texts for either cover are too voluminous to fit legibly, you should put the first ones listed (as many as fit reasonably) on the actual cover, and continue the rest onto adjacent pages.

If you publish or distribute Opaque copies of the Document numbering more than 100, you must either include a machine-readable Transparent copy along with each Opaque copy, or state in or with each Opaque copy a computer-network location from which the general network-using public has access to download using public-standard network protocols a complete Transparent copy of the Document, free of added material. If you use the latter option, you must take reasonably prudent steps, when you begin distribution of Opaque copies in quantity, to ensure that this Transparent copy will remain thus accessible at the stated location until at least one year after the last time you distribute an Opaque copy (directly or through your agents or retailers) of that edition to the public.

It is requested, but not required, that you contact the authors of the Document well before redistributing any large number of copies, to give them a chance to provide you with an updated version of the Document.

MODIFICATIONS

You may copy and distribute a Modified Version of the Document under the conditions of sections 2 and 3 above, provided that you release the Modified Version under precisely this License, with the Modified Version filling the role of the Document, thus licensing distribution and modification of the Modified Version to whoever possesses a copy of it. In addition, you must do these things in the Modified Version:

A. Use in the Title Page (and on the covers, if any) a title distinct from that of the Document, and from those of previous versions (which should, if there were any, be listed in the History section of the Document). You may use the same title as a previous version if the original publisher of that version gives permission.

B. List on the Title Page, as authors, one or more persons or entities responsible for authorship of the modifications in the Modified Version, together with at least five of the principal authors of the Document (all of its principal authors, if it has fewer than five), unless they release you from this requirement.

C. State on the Title page the name of the publisher of the Modified Version, as the publisher.

D. Preserve all the copyright notices of the Document.

E. Add an appropriate copyright notice for your modifications adjacent to the other copyright notices.

F. Include, immediately after the copyright notices, a license notice giving the public permission to use the Modified Version under the terms of this License, in the form shown in the Addendum below.

G. Preserve in that license notice the full lists of Invariant Sections and required Cover Texts given in the Document's license notice.

H. Include an unaltered copy of this License.

I. Preserve the section Entitled "History", Preserve its Title, and add to it an item stating at least the title, year, new authors, and publisher of the Modified Version as given on the Title Page. If there is no section Entitled "History" in the Document, create one stating the title, year, authors, and publisher of the Document as given on its Title Page, then add an item describing the Modified Version as stated in the previous sentence.

J. Preserve the network location, if any, given in the Document for public access to a Transparent copy of the Document, and likewise the network locations given in the Document for previous versions it was based on. These may be placed in the "History" section. You may omit a network location for a work that was published at least four years before the Document itself, or if the original publisher of the version it refers to gives permission.

K. For any section Entitled "Acknowledgements" or "Dedications", Preserve the Title of the section, and preserve in the section all the substance and tone of each of the contributor acknowledgements and/or dedications given therein.

L. Preserve all the Invariant Sections of the Document, unaltered in their text and in their titles. Section numbers or the equivalent are not considered part of the section titles.

M. Delete any section Entitled "Endorsements". Such a section may not be included in the Modified Version.

N. Do not retitle any existing section to be Entitled "Endorsements" or to conflict in title with any Invariant Section.

O. Preserve any Warranty Disclaimers.

If the Modified Version includes new front-matter sections or appendices that qualify as Secondary Sections and contain no material copied from the Document, you may at your option designate some or all of these sections as invariant. To do this, add their titles to the list of Invariant Sections in the Modified Version's license notice. These titles must be distinct from any other section titles.

You may add a section Entitled "Endorsements", provided it contains nothing but endorsements of your Modified Version by various parties--for example, statements of peer review or that the text has been approved by an organization as the authoritative definition of a standard.

You may add a passage of up to five words as a Front-Cover Text, and a passage of up to 25 words as a Back-Cover Text, to the end of the list of Cover Texts in the Modified Version. Only one passage of Front-Cover Text and one of Back-Cover Text may be added by (or through arrangements made by) any one entity. If the Document already includes a cover text for the same cover, previously added by you or by arrangement made by the same entity you are acting on behalf of, you may not add another; but you may replace the old one, on explicit permission from the previous publisher that added the old one.

The author(s) and publisher(s) of the Document do not by this License give permission to use their names for publicity for or to assert or imply endorsement of any Modified Version.

COMBINING DOCUMENTS

You may combine the Document with other documents released under this License, under the terms defined in section 4 above for modified versions, provided that you include in the combination all of the Invariant Sections of all of the original documents, unmodified, and list them all as Invariant Sections of your combined work in its license notice, and that you preserve all their Warranty Disclaimers.

The combined work need only contain one copy of this License, and multiple identical Invariant Sections may be replaced with a single copy. If there are multiple Invariant Sections with the same name but different contents, make the title of each such section unique by adding at the end of it, in parentheses, the name of the original author or publisher of that section if known, or else a unique number. Make the same adjustment to the section titles in the list of Invariant Sections in the license notice of the combined work.

In the combination, you must combine any sections Entitled “History” in the various original documents, forming one section Entitled “History”; likewise combine any sections Entitled “Acknowledgements”, and any sections Entitled “Dedications”. You must delete all sections Entitled “Endorsements”.

COLLECTIONS OF DOCUMENTS

You may make a collection consisting of the Document and other documents released under this License, and replace the individual copies of this License in the various documents with a single copy that is included in the collection, provided that you follow the rules of this License for verbatim copying of each of the documents in all other respects.

You may extract a single document from such a collection, and distribute it individually under this License, provided you insert a copy of this License into the extracted document, and follow this License in all other respects regarding verbatim copying of that document.

AGGREGATION WITH INDEPENDENT WORKS

A compilation of the Document or its derivatives with other separate and independent documents or works, in or on a volume of a storage or distribution medium, is called an “aggregate” if the copyright resulting from the compilation is not used to limit the legal rights of the compilation’s users beyond what the individual works permit. When the Document is included in an aggregate, this License does not apply to the other works in the aggregate which are not themselves derivative works of the Document.

If the Cover Text requirement of section 3 is applicable to these copies of the Document, then if the Document is less than one half of the entire aggregate, the Document’s Cover Texts may be placed on covers that bracket the Document within the aggregate, or the electronic equivalent of covers if the Document is in electronic form. Otherwise they must appear on printed covers that bracket the whole aggregate.

TRANSLATION

Translation is considered a kind of modification, so you may distribute translations of the Document under the terms of section 4. Replacing Invariant Sections with translations requires special permission from their copyright holders, but you may include translations of some or all Invariant Sections in addition to the original versions of these Invariant Sections. You may include a translation of this License, and all the license notices in the Document, and any Warranty Disclaimers, provided that you also include the original English version of this License and the original versions of those notices and disclaimers. In case of a disagreement between the translation and the original version of this License or a notice or disclaimer, the original version will prevail.

If a section in the Document is Entitled “Acknowledgements”, “Dedications”, or “History”, the requirement (section 4) to Preserve its Title (section 1) will typically require changing the actual title.

TERMINATION

You may not copy, modify, sublicense, or distribute the Document except as expressly provided for under this License. Any other attempt to copy, modify, sublicense or distribute the Document is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.

FUTURE REVISIONS OF THIS LICENSE

The Free Software Foundation may publish new, revised versions of the GNU Free Documentation License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns. See http://www.gnu.org/copyleft/.

Each version of the License is given a distinguishing version number. If the Document specifies that a particular numbered version of this License “or any later version” applies to it, you have the option of following the terms and conditions either of that specified version or of any later version that has been published (not as a draft) by the Free Software Foundation. If the Document does not specify a version number of this License, you may choose any version ever published (not as a draft) by the Free Software Foundation.

ADDENDUM: How to use this License for your documents

To use this License in a document you have written, include a copy of the License in the document and put the following copyright and license notices just after the title page:

```
Copyright (c) YEAR YOUR NAME.
Permission is granted to copy, distribute and/or modify this document
under the terms of the GNU Free Documentation License, Version 1.2
or any later version published by the Free Software Foundation;
with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts.
A copy of the license is included in the section entitled “GNU
Free Documentation License”.
```

If you have Invariant Sections, Front-Cover Texts and Back-Cover Texts, replace the “with...Texts.” line with this:

```
with the Invariant Sections being LIST THEIR TITLES, with the
Front-Cover Texts being LIST, and with the Back-Cover Texts being LIST.
```

If you have Invariant Sections without Cover Texts, or some other combination of the three, merge those two alternatives to suit the situation.

If your document contains nontrivial examples of program code, we recommend releasing these examples in parallel under your choice of free software license, such as the GNU General Public License, to permit their use in free software.